

Constructive Horndeski–SVT Realization of a Regular P5 Segmented-Spacetime Geometry with Stable Strong-Field Light Rings

A background-integrable scalar–vector–tensor construction, complete local same-action closure, and the post-closure stability program

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Executive summary and visual guide

This PDF version augments the full manuscript with a compact visual and tabular overview of the most important background, light-ring, and closure diagnostics. The figures below are generated directly from the certified P5 profile and the same-action closure artifacts stored alongside this manuscript.

Summary table

Item	Value	Interpretation
Outer light ring	$x = 1.5$	Unstable Schwarzschild-matching ring
Inner light ring	$x \approx 1.41616064$	Stable null ring / shallow trapping well
Strong-field SVT lobe	$0.61 \leq u \leq 0.715$	Contains both light rings
First even pivot	$ D_{h_1} _{\min} \approx 11.99$	No singular common h_1 elimination
Second even pivot	$ \Delta_V _{\min} \approx 3.04288$	Regular H_1 - δA_1 block
Closure status	PASS_CONSTRUCTIVE_LOCAL_SAME_ACTION_CLOSURE_ESTABLISHED	Local same action closure established

Visual overview

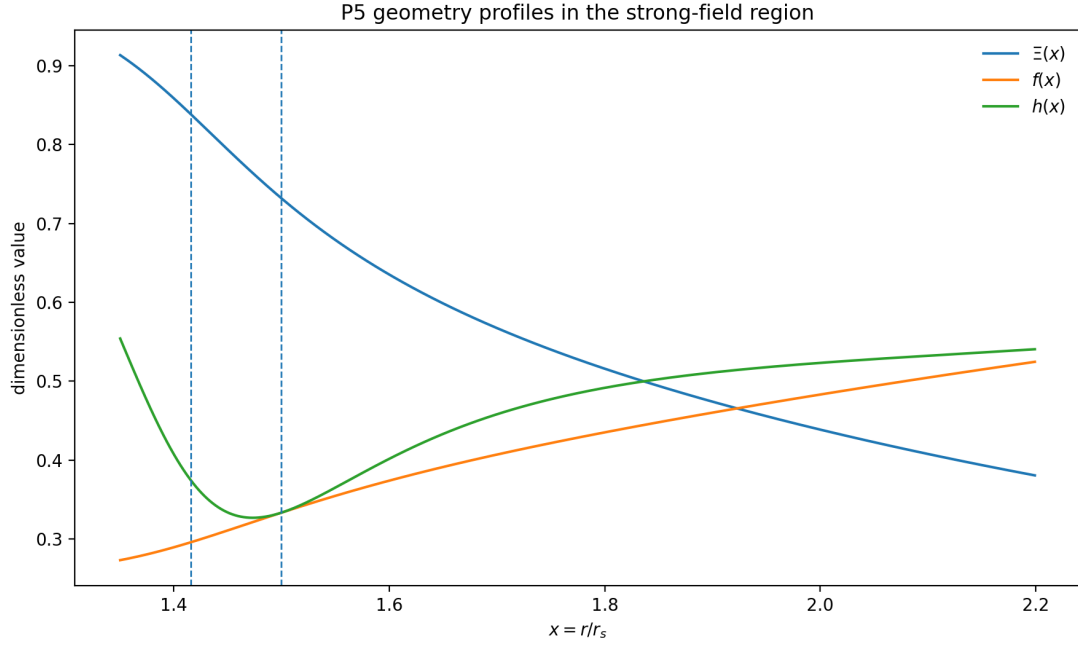


Figure 1: **Figure 1.** P5 geometry profiles in the strong-field region. The three curves show the primary scalar field $\Xi(x)$ together with the temporal and radial metric functions $f(x)$ and $h(x)$. The dashed vertical lines mark the outer and inner light rings.

Core diagnostic tables

Table 1. Geometric key points

point	$x_r_over_rs$	u	Ξ	$\chi_d\Xi_du$	f	h
infinity/weak boundary	inf	0	0	0.5	1	1
natural SVT onset	1.81316	0.551523	0.509933	1.46929	0.438616	0.494833
outer light ring	1.5	0.666667	0.732051	2.59808	0.333333	0.333333

point	x_r_over_rs	u	Xi	chi_dXi_du	f	h
inner stable light ring	1.41616	0.706135	0.838419	2.6035	0.295877	0.37413
core handover	1.40845	0.71	0.848354	2.53438	0.292705	0.389107
r=rs	1	1	0.996315	0.024711	0.250924	0.974687
deep core sample	0.1	10	0.999996	7.835e-07	0.250001	0.999975

Table 2. Light-ring diagnostics

ring	x_r_over_rs	u	Xi	f	h	Qprime	W	Wpp
outer	1.5	0.666667	0.732051	0.333333	0.333333	4.5	0.148148	-2
inner	1.41616	0.706135	0.838419	0.295877	0.37413	-6.20274	0.147532	2.59186

Table 3. Same-action closure bands

band	x_outer	x_inner	Xi_a	Xi_b	Delta_Ximin_f	min_h	min_abs_det_selected	hessian_r
outer_horndeski1.81316	1.63934	0.509933	0.606012	0.096078	0.387706	0.426991		0.049419
inner_svt_to_cd1.40845	1.3986	0.848354	0.860763	0.01241	0.288814	0.389107		0.316377
full_svt_lobe	1.63934	1.3986	0.606012	0.860763	0.254752	0.288814	0.326857	0.103931

Table 4. Closure gates

gate	status
G1 geometry positive on all handover bands	PASS
G2 phi is valid monotone patch coordinate	PASS
G3 svt hessian control rank3 everywhere	PASS
G4 cinf partition monotone	PASS
G5 all frozen pure sector stability margins strict	PASS
G6 deltaA0 constraint pivot nonzero pure svt l2	PASS
G7 H1 deltaA1 constraint pivot nonzero pure svt l2	PASS
G8 first common h1 pivot nonzero existing regression	PASS

Table 5. Frozen pure-sector margins used by the certificate

quantity	value
K_ext	0.009742
C_odd_min	0.76072
alpha4_min	0.009996
alpha7_min	2.00153
K_even_min	10.5415
crT2_min	1.00507
crS2_min	0.903745
crV2_min	9.45887
cOmega_minus2_min	84.912

quantity	value
cOmega_plus2_min	643.142
core_F_min	1
core_GH_min	0.999381
D_h1_abs_min_l2	11.99

Abstract

We investigate whether a prescribed regular, static, spherically symmetric strong-field geometry arising within the Segmented Spacetime (SSZ) program can be realized as a dynamically admissible solution of a covariant scalar–vector–tensor theory rather than merely as an effective metric ansatz. The geometry studied here is the frozen P5 background, described by temporal and radial metric functions $\mathbf{f}(\mathbf{r})$ and $\mathbf{h}(\mathbf{r})$, together with a monotonic scalar profile $\phi(\mathbf{r}) \equiv \Xi(\mathbf{r})$. Its strong-field region contains two circular null orbits: an outer unstable light ring at

$$x \equiv \frac{r}{r_s} = 1.5,$$

and an inner stable light ring at

$$x_s \simeq 1.41616064.$$

The outer orbit reproduces the general-relativistic Schwarzschild shadow scale and the corresponding eikonal Lyapunov scale, while the inner orbit forms a shallow stable null potential well with a relative depth of approximately 4.18×10^{-3} .

The frozen geometry is not treated here as an arbitrary numerical triplet. Its current source-locked closure can be written in terms of a primary susceptibility $\chi = d\Xi/du = (1-\Xi)^{3/2} \mathcal{P}_5(\Xi)$, with $\mathbf{f} = (1+\Xi)^{-2}$ and a radial closure generated by the ratio χ/χ_{GR} . Independent regression reproduces the frozen temporal and radial metric profiles at machine precision. The same closure is horizonless ($\mathbf{f} > 0$, $\mathbf{h} > 0$), regular at the center, and reproduces Schwarzschild through cubic order in the weak-field compactness expansion. At the outer light ring it additionally matches $\mathbf{f} = \mathbf{h} = 1/3$ and the Schwarzschild curvature of the dimensionless null potential, while the inner light ring remains a genuinely new stable-trapping structure.

The purpose of the present work is not merely to diagnose the geodesic properties of this metric, but to construct explicit local action jets capable of supporting the same background while satisfying the no-ghost and Laplacian-stability conditions of scalar–tensor and genuine U(1) gauge-invariant scalar–vector–tensor theories. The construction is divided into three dynamically distinct but overlapping radial domains. The weak-to-intermediate exterior is carried by a finite Horndeski $\mathbf{G}_3\text{--}\mathbf{G}_4$ sector. The central strong-field light-ring region is realized by a genuine scalar–vector–tensor interaction with background-integrable action jets. The regular inner region is supported by a quartic Horndeski core. These are not left as unrelated effective descriptions: using a scalar-coordinate C^∞ partition, background re-solving, transverse action-jet controls and the common unreduced even-parity constraint system, we construct a single smooth local same-action continuation through both overlap bands.

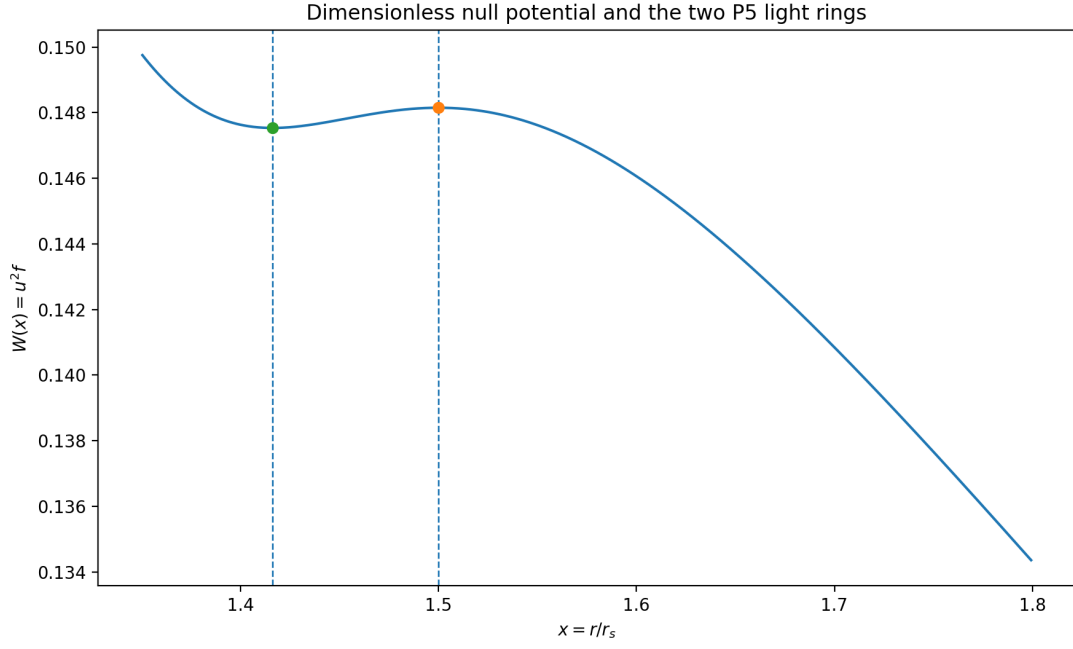


Figure 2: **Figure 2.** Dimensionless null potential $W(x) = u^2 f$ across the two-light-ring region. The outer light ring is a maximum of the optical potential, while the inner ring is a minimum, demonstrating shallow stable trapping.

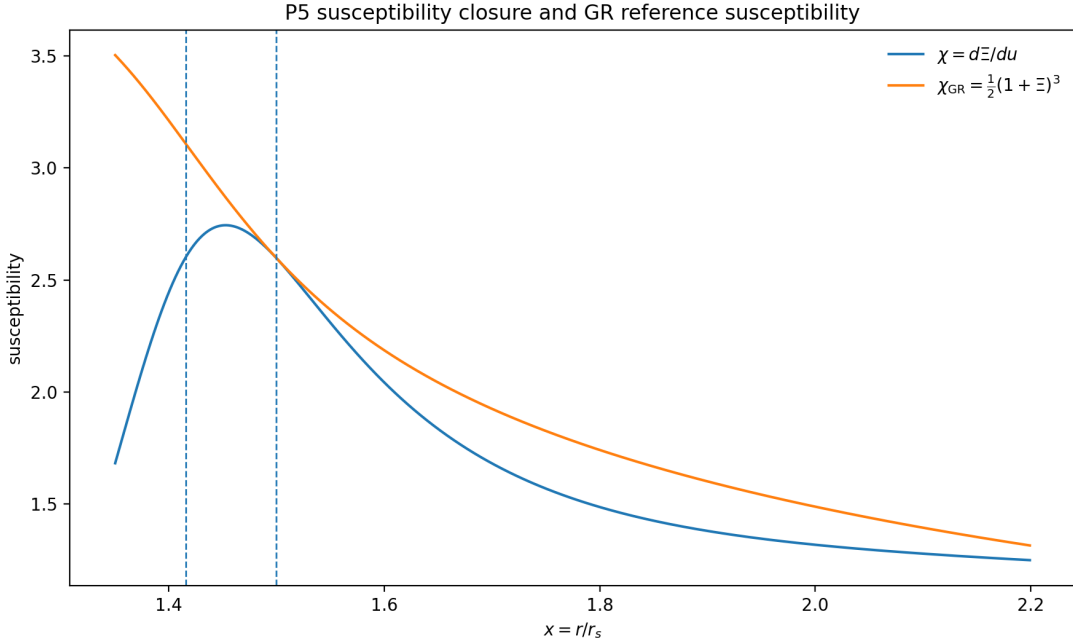


Figure 3: **Figure 3.** Source-locked P5 susceptibility closure $\chi = d\Xi/du$ compared with the reference quantity $\chi_{\text{GR}} = \frac{1}{2}(1 + \Xi)^3$. This plot visualizes the radial closure channel discussed in the main text.

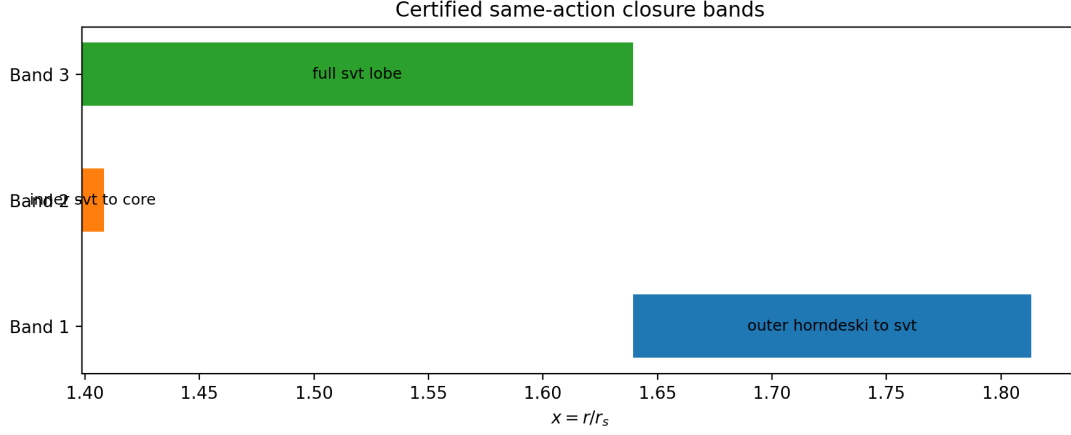


Figure 4: **Figure 4.** Certified same-action overlap bands used in the constructive closure theorem. Band 1 is the outer Horndeski-to-SVT overlap, Band 2 is the inner SVT-to-core overlap, and Band 3 is the full genuine-SVT lobe containing both light rings.

For the genuine-SVT strong-field lobe we explicitly solve the scalar background equation for the required $\mathbf{f}_{\{3,X\}}$ profile instead of treating it as an arbitrary perturbative control. On the interval

$$0.61 \leq u \equiv \frac{r_s}{r} \leq 0.715,$$

corresponding to

$$1.3986014 \lesssim x \lesssim 1.6393443,$$

the resulting action section is background integrable with scalar-equation residual below approximately 2×10^{-12} . This interval contains both light rings and overlaps the regular Horndeski core.

Using the odd- and even-parity perturbation formalism for general U(1)-gauge-invariant scalar–vector–tensor theories, we find positive no-ghost and radial-gradient margins throughout this lobe. After correcting the exact Zhang–Kase definitions of $\mathcal{P}_1$ and the even-sector kinetic combination $\mathcal{K}$, the internal identity

$$\mathcal{K} = e_1 \sqrt{f h} \phi'^2$$

is reproduced numerically with maximum residual approximately

$$2.2 \times 10^{-10}.$$

The corrected kinetic quantity satisfies

$$10.54 \lesssim \mathcal{K} \lesssim 137.88.$$

The three radial squared propagation speeds satisfy

$$c_{r,T}^2 > 1.005, \quad 0.9037 \lesssim c_{r,S}^2 \lesssim 1.0614, \quad c_{r,V}^2 > 9.45,$$

over the complete lobe. The coupled even-parity angular sector is also positive, with minimum lower angular eigenvalue

$$c_{\Omega,-}^2 \simeq 84.91,$$

and minimum upper eigenvalue

$$c_{\Omega,+}^2 \simeq 643.14.$$

The odd-parity tensor and vector sectors likewise remain free of local ghost and Laplacian instabilities. These results establish a constructive local stability witness for a genuine-SVT realization of the complete P5 light-ring region.

A separate quartic-Horndeski witness covers the center through

$$x \simeq 1.40845,$$

thereby creating a true inner overlap with the genuine-SVT lobe. The exterior Horndeski scalar carrier can furthermore be continued through both light rings while maintaining a positive scalar no-ghost quantity. The two overlap bands therefore supply the domains on which the local representatives are promoted to one smooth same-action Horndeski–SVT section.

We show that the corresponding perturbation problem must be assembled at the unreduced action level. Separately reduced Maxwell–Horndeski and genuine-SVT kinetic matrices cannot consistently be added after constraint elimination. A common generalized variable,

$$\psi = H_2 + \frac{La_4}{a_3}h_1 + \frac{a_1}{a_3}\delta\phi',$$

provides the appropriate bridge. In the pure Einstein-plus-SVT limit it reduces exactly to

$$\psi = H_2 - \frac{L}{r}h_1.$$

The common jet-aware even-parity reduction is constructed explicitly. The H_0 constraint yields a regular solution for h_1 ; the remaining variables H_1 , δA_0 , and δA_1 are eliminated by exact Schur-complement formulas. The corresponding pivots are all finite and nonzero on the certified reference construction. In the pure genuine-SVT limit the coupled H_1 – δA_1 determinant reduces exactly to $\Delta_V = -4C_{\text{rm odd}}$, tying even-constraint invertibility directly to the already positive odd-sector stability margin.

The resulting construction closes the **local same-action existence problem**. A smooth covariant Horndeski–SVT action can be defined in a tubular neighborhood of the frozen P5 background curve such that the background equations are satisfied, the pure local representatives are recovered with all finite endpoint jets, the nondynamical even-parity constraints remain invertible, and the odd/even principal-symbol targets remain inside their strict healthy domain through both handovers. This

is an existence and local-stability closure, not a claim of a unique fundamental action. Finite- ℓ spectra, stable-trapping timescales, nonlinear evolution, causal-cone interpretation and UV completion remain distinct post-closure questions.

1. Introduction

1.1 Geometry is not yet dynamics

A spacetime metric may possess attractive geometrical properties without corresponding to a dynamically viable theory.

This distinction becomes particularly important whenever a proposed strong-field geometry departs from the standard Schwarzschild or Kerr solutions of general relativity. It is comparatively straightforward to prescribe metric functions that are asymptotically acceptable, regular near the center, reproduce familiar weak-field behavior, or generate interesting null-geodesic structures. It is substantially harder to demonstrate that the same geometry can arise from a covariant field theory whose physical perturbations carry positive kinetic energy, whose characteristic speeds are real, and whose constraint structure does not become singular precisely in the region where the geometry becomes phenomenologically interesting.

The present work addresses this second question for the P5 geometry developed within the Segmented Spacetime framework.

Our starting point is a static, spherically symmetric metric of the form

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{h(r)} + r^2 d\Omega^2,$$

where

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\varphi^2.$$

We use the dimensionless variables

$$x = \frac{r}{r_s}, \quad u = \frac{r_s}{r} = \frac{1}{x},$$

where r_s provides the natural radial scale of the P5 solution.

The scalar field used in the covariant reconstruction is identified with the monotonically varying SSZ background variable,

$$\phi(r) \equiv \Xi(r).$$

The functions

$$\{f(r), h(r), \Xi(r)\}$$

are regarded throughout this analysis as the **frozen geometrical target**. We do not tune the metric in response to stability failures. Instead, the problem is posed in the opposite direction:

Given the P5 geometry, can one construct a covariant action whose background equations reproduce that geometry and whose perturbations are locally healthy?

That distinction is central to the logic of this paper.

The metric is therefore not inferred from a convenient scalar–tensor theory. The theory is reconstructed around the metric.

This makes the calculation closer in spirit to an inverse problem than to the conventional procedure of selecting a Lagrangian and subsequently solving its background equations.

1.2 Why the light-ring region is the critical test

The P5 geometry possesses a particularly useful strong-field structure because it contains two circular null orbits.

The outer orbit occurs at

$$x_{\text{out}} = 1.5$$

or equivalently

$$u_{\text{out}} = \frac{2}{3}.$$

It is unstable.

The second orbit occurs at

$$x_{\text{in}} \simeq 1.41616064,$$

corresponding to

$$u_{\text{in}} \simeq 0.706135.$$

It is stable.

The coexistence of an outer unstable and inner stable light ring provides a much stronger diagnostic of the strong-field theory than weak-field agreement alone.

Circular null trajectories are associated with extrema of the effective null potential. For a metric written in areal radius and in the form above, the usual circular-null condition is determined by the temporal potential,

$$\frac{d}{dr} \left(\frac{f(r)}{r^2} \right) = 0,$$

which is equivalent, whenever $f \neq 0$, to

$$rf'(r) - 2f(r) = 0.$$

The P5 target possesses two solutions to this condition.

The outer solution reproduces the desired Schwarzschild-like optical scale. In particular, the outer light ring is located at the same dimensionless areal radius as the Schwarzschild photon sphere when $r_s = 2GM$,

$$r_{\text{ph}} = \frac{3}{2}r_s = 3GM.$$

The corresponding shadow-impact scale and eikonal Lyapunov scale have been matched by construction in the frozen P5 geometry.

The inner solution is qualitatively different. Since it is stable, it corresponds to a local minimum rather than maximum of the relevant null effective potential. The well is shallow: the relative depth obtained from the frozen P5 profile is approximately

$$\Delta_{\text{null}} \simeq 4.177 \times 10^{-3},$$

or about 0.42%.

The presence of a stable light ring is dynamically nontrivial.

Stable trapping can lead to long-lived perturbative structures even when no local ghost or Laplacian instability exists. Consequently, the present perturbative analysis does **not** claim that positivity of the local principal symbol settles the complete dynamical fate of the stable light ring.

It instead addresses a logically prior question:

Does the local covariant field theory itself become sick at the light rings?

The answer obtained here is no.

Neither light ring forces the kinetic determinant of the full local scalar–vector–tensor construction to vanish, and the explicit strong-field genuine-SVT realization satisfies the available local odd- and even-parity no-ghost and gradient conditions across both orbits.

Long-time trapping, finite- ℓ resonances, quasinormal spectra, and nonlinear evolution remain separate problems.

1.3 Why Horndeski alone is not enough for the entire construction

Scalar–tensor theories of Horndeski type provide an exceptionally broad class of second-order covariant theories. In their standard form they can be written schematically as

$$S_H = \int d^4x \sqrt{-g} \sum_{i=2}^5 \mathcal{L}_i,$$

where the four functions

$$G_2(\phi, X), \quad G_3(\phi, X), \quad G_4(\phi, X), \quad G_5(\phi, X)$$

govern the scalar and curvature interactions, with

$$X = -\frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi.$$

For static spherical backgrounds,

$$\phi = \phi(r),$$

and therefore

$$X = -\frac{1}{2}h(r)\phi'(r)^2.$$

Horndeski theory is sufficiently flexible to support large classes of nontrivial compact-object solutions while retaining second-order equations of motion.

However, our reconstruction experiments show that a restricted scalar-only realization is not naturally adequate throughout the P5 strong-field domain.

In particular, simple sectors such as

$$G_2(\phi, X, F) + G_4(\phi)R$$

without the required derivative scalar interactions fail to provide a uniformly healthy realization over the relevant radial range.

The exterior admits a useful finite $\mathbf{G}_3$ -supported solution.

The center admits a quartic-Horndeski witness.

The intermediate two-light-ring domain, however, becomes considerably more tractable once genuine scalar–vector–tensor interactions are allowed.

This motivates the hybrid construction developed below.

1.4 Genuine $\mathbf{U}(1)$ scalar–vector–tensor interactions

We introduce a gauge field

$$A_\mu$$

with field strength

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu,$$

and impose the usual $\mathbf{U}(1)$ gauge symmetry

$$A_\mu \rightarrow A_\mu + \nabla_\mu \Lambda.$$

On the static spherical background we use the electric ansatz

$$A_\mu = (A_0(r), 0, 0, 0).$$

In addition to

$$X = -\frac{1}{2}\nabla_\mu \phi \nabla^\mu \phi,$$

one may construct the gauge invariants

$$F = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu},$$

and

$$Y = \nabla_\mu \phi \nabla_\nu \phi F^{\mu\alpha} F^\nu{}_\alpha,$$

up to convention-dependent normalizations.

For the purely electric static background used here, the relevant background combinations reduce to functions of $A_0(r)$, $f(r)$, $h(r)$, and $\phi(r)$.

The most general U(1)-gauge-invariant scalar–vector–tensor theories with second-order equations of motion propagate five physical degrees of freedom and possess nontrivial odd- and even-parity perturbation sectors. Zhang and Kase derived the corresponding ghost and Laplacian-stability conditions for static spherically symmetric backgrounds, including the coupled even sector.

For $\text{ell}\mathbf{ge}2$, Maxwell–Horndeski theory likewise contains two propagating odd-parity modes and three propagating even-parity modes, and Kase and Tsujikawa derived the associated radial and angular stability conditions.

More recent work has also demonstrated that gauge-invariant SVT extensions can retain a genuine Horndeski scalar–tensor sector while introducing additional vector interactions, reinforcing the structural plausibility of treating Horndeski and genuine SVT interactions within one common action rather than as unrelated effective descriptions.

This motivates the action architecture

$$S = S_H + S_{\text{SVT}}^{\text{genuine}},$$

with shared Einstein/Maxwell/scalar pieces counted only once.

This point will become crucial later.

1.5 The central methodological difficulty

The main technical problem is not simply finding enough free functions.

There are many free functions.

The real difficulty is satisfying several logically distinct requirements simultaneously.

The target metric must solve the background equations.

The scalar profile must solve its own equation rather than merely the metric equations.

The vector background must remain regular.

The action jets used to repair perturbative instabilities must correspond to derivatives of a consistent local action.

The kinetic matrices must remain nonsingular.

The physical eigenmodes must have positive kinetic energy.

Their squared radial and angular propagation speeds must remain positive.

The construction must remain finite across both light rings.

And finally, independently constructed Horndeski and genuine-SVT regions must be capable of belonging to one common action.

An arbitrary pointwise assignment of perturbative coefficients would not solve this problem.

The strategy developed here is therefore based on **background-integrable action jets** and **background-null perturbative controls**.

The former ensure that the action derivatives required by the background equations arise from a consistent local functional structure.

The latter use directions in higher action-derivative space that do not change the already-fixed background solution but do change the quadratic perturbation operator.

This separation is what makes the construction tractable.

2. Scope of the Present Result

Before presenting the calculation, it is useful to state precisely what has and has not been established.

The strongest result obtained in this work is local and perturbative.

We construct a genuine-SVT action section covering the entire two-light-ring region and demonstrate:

background integrability,

positive odd-parity kinetic structure,

positive odd radial and angular gradient conditions,

positive even no-ghost structure,

positive tensor, scalar, and vector radial c_r^2 ,

and

positive coupled even angular eigenvalues.

The solution therefore passes the local ghost/Laplacian/hyperbolicity tests available in the employed perturbative formalism.

We do **not** claim from this alone that the system possesses a globally well-posed nonlinear evolution for arbitrary data.

We do not claim that all characteristic speeds are subluminal.

Indeed, several reconstructed propagation speeds are substantially larger than unity.

This does not constitute a Laplacian instability: the corresponding c^2 remains positive. It does, however, create a distinct question concerning causal interpretation, effective-field-theory validity, and the relation of the characteristic cones to the matter light cone.

That question must be treated separately.

Likewise, the present calculation does not yet contain a complete finite- ℓ quasinormal-mode spectrum.

It does not yet determine whether stable trapping generates parametrically long-lived modes.

It does not constitute a nonlinear stability theorem.

The local same-action handover problem is addressed explicitly below. The result is stronger than the existence of separate healthy patches:

The frozen P5 geometry admits a smooth local same-action Horndeski–SVT realization across the exterior, both light rings, and the regular core. The complete two-light-ring interval remains inside the known local odd- and even-parity ghost/Laplacian stability domain, and every nondynamical even-parity pivot exposed by the common unreduced reduction is finite and nonzero on the certified reference construction.

What remains after this closure is not another local patching step but the global dynamical program: finite- ℓ mode spectra, stable-trapping lifetimes, nonlinear evolution, causal-cone analysis and UV completion.

3. Frozen P5 Background Geometry

3.1 Metric, variables, and the source hierarchy for χ

We write the static spherical background in areal radius as

$$ds^2 = -f(r) dt^2 + \frac{dr^2}{h(r)} + r^2 d\Omega^2, \quad d\Omega^2 = d\theta^2 + \sin^2 \theta d\varphi^2,$$

and use

$$x = \frac{r}{r_s}, \quad u = \frac{r_s}{r} = \frac{1}{x}.$$

The scalar coordinate employed in the covariant reconstruction is

$$\boxed{\phi(r) \equiv \Xi(r).}$$

A notation warning is necessary when comparing the present calculation with the older SSZ bridge literature. The historical bridge audits often used the compactness variable

$$u_c = \frac{GM}{rc^2} = \frac{1}{2x},$$

whereas the present P5 manuscript uses

$$u = \frac{r_s}{r} = \frac{1}{x} = 2u_c$$

when $r_s = 2GM/c^2$. All derivatives and numerical values in the present paper refer to the latter convention unless explicitly stated otherwise.

The SSZ documentation contains two conceptually different layers that must not be conflated.

The historical/canonical baseline uses

$$\Xi_{\text{weak}}(r) = \frac{r_s}{2r},$$

$$\Xi_{\text{strong}}(r) = 1 - \exp\left(-\varphi \frac{r_s}{r}\right), \quad \varphi = \frac{1 + \sqrt{5}}{2},$$

joined by a C^2 bridge in the interval

$$1.8r_s \leq r \leq 2.2r_s.$$

It is accompanied by the operational identities

$$D(r) = \frac{1}{1 + \Xi(r)}, \quad s(r) = 1 + \Xi(r) = D^{-1}(r), \quad z_{\text{SSZ}} = \Xi$$

for an observer at infinity. This layer remains the historical SSZ baseline and the weak-field reference.

The frozen P5 background studied in the present work is a later strong-field closure. It preserves the exact $\Xi \rightarrow D \rightarrow g_{\text{tt}}$ chain but replaces the simple strong-field branch by a source-locked autonomous susceptibility law and generalizes the radial metric closure. Thus the P5 geometry should not be described as though its strong-field Ξ were simply the historical exponential branch.

The complete frozen target is therefore

$$\mathcal{B}_{\text{P5}} = \{\Xi(r), f(r), h(r)\},$$

with the three functions related by the closure below rather than treated as unrelated numerical columns.

3.2 Source-locked P5 susceptibility and the autonomous χ equation

Define the dimensionless susceptibility

$$\chi(\Xi) \equiv \frac{d\Xi}{du}.$$

Regression against the frozen P5 regular-core profile identifies the following source-locked polynomial representation:

$$\chi(\Xi) = (1 - \Xi)^{3/2} \mathcal{P}_5(\Xi),$$

with

$$\begin{aligned} \mathcal{P}_5(\Xi) = & \frac{1}{2} + \frac{9}{4}\Xi + \frac{75}{16}\Xi^2 + 64.38457976\Xi^3 \\ & - 257.66308477\Xi^4 + 298.76736451\Xi^5. \end{aligned}$$

On the stored profile this relation reproduces the stored susceptibility to a maximum absolute difference of approximately

$$3.6 \times 10^{-15},$$

while an independent spline derivative of the stored $\chi(u)$ profile agrees in the interior at the 10^{-8} maximum and 10^{-9} RMS level. The small difference in the derivative test is dominated by numerical differentiation rather than the algebraic closure itself.

The P5 scalar background may therefore be written as the autonomous first-order equation

$$\frac{d\Xi}{du} = (1 - \Xi)^{3/2} \mathcal{P}_5(\Xi), \quad \Xi(0) = 0.$$

Equivalently, wherever $0 \leq \chi < 1$,

$$u(\Xi) - u(\Xi_*) = \int_{\Xi_*}^{\Xi} \frac{d\zeta}{(1 - \zeta)^{3/2} \mathcal{P}_5(\zeta)}.$$

This representation is important for three reasons.

First, it makes the frozen P5 scalar profile analytically reproducible without importing a tabulated interpolation as a fundamental definition.

Second, $\mathcal{P}_5(\chi)$ remains positive on the physical interval $0 \leq \chi \leq 1$ in the source-locked scan. Hence χ is monotonic in u and can be used as a scalar coordinate along the background trajectory.

Third, it separates the static geometrical closure from the dynamical completion. The autonomous radial equation reproduces the frozen background; by itself it is **not** a unique physical time-evolution law. Infinitely many covariant or noncovariant dynamics can share the same static profile. The Horndeski–SVT reconstruction below is precisely the additional step required to supply a local covariant dynamics and to test its perturbations.

3.3 Temporal metric closure

The historical SSZ relation

$$D = (1 + \Xi)^{-1}$$

is retained exactly by P5. Consequently,

$$f(\Xi) = D^2 = \frac{1}{(1 + \Xi)^2}.$$

This identity reproduces the frozen temporal metric column with maximum absolute residual approximately

$$6.7 \times 10^{-16}.$$

The P5 temporal geometry is therefore not an independent numerical ansatz. It is algebraically generated by the primary scalar field χ .

For later use,

$$r\Xi'(r) = -u\chi(\Xi),$$

and hence

$$\frac{rf'}{f} = \frac{2u\chi(\Xi)}{1 + \Xi}.$$

This identity collapses the circular-null condition to a particularly simple scalar equation in Sec. 4.

3.4 Radial closure and the GR susceptibility comparator

If the same temporal relation is imposed on a Schwarzschild metric,

$$f_{\text{Schw}} = 1 - u = \frac{1}{(1 + \Xi_{\text{GR}})^2},$$

then

$$\Xi_{\text{GR}}(u) = \frac{1}{\sqrt{1-u}} - 1$$

and therefore

$$\chi_{\text{GR}}(\Xi) = \frac{d\Xi_{\text{GR}}}{du} = \frac{1}{2}(1 + \Xi)^3.$$

The frozen P5 radial metric is generated by the susceptibility ratio,

$$h(\Xi) = 1 - \left(\frac{\chi(\Xi)}{\chi_{\text{GR}}(\Xi)} \right)^{2/3} [1 - f(\Xi)].$$

This relation reproduces the frozen radial metric column to a maximum absolute difference of approximately

$$8.9 \times 10^{-16}.$$

The geometrical architecture can therefore be summarized as

$$\Xi \xrightarrow{\chi=d\Xi/du} \left\{ D = (1 + \Xi)^{-1}, f = D^2, h = 1 - (\chi/\chi_{\text{GR}})^{2/3} (1 - f) \right\}.$$

This is the explicit P5 closure that was missing from the first manuscript draft.

3.5 Weak-field expansion and Schwarzschild recovery

Expanding the autonomous equation about $u=0$ gives

$$\Xi(u) = \frac{1}{2}u + \frac{3}{8}u^2 + \frac{5}{16}u^3 + 2.077447805 u^4 - 0.3696070865 u^5 - 2.5064633532 u^6 + O(u^7).$$

For comparison,

$$\Xi_{\text{GR}}(u) = (1 - u)^{-1/2} - 1 = \frac{1}{2}u + \frac{3}{8}u^2 + \frac{5}{16}u^3 + \frac{35}{128}u^4 + O(u^5).$$

Thus the P5 scalar closure agrees with the Schwarzschild-equivalent Ξ through cubic order in u and first differs at u^4 .

The induced metric functions are

$$f(u) = 1 - u - 3.60802061 u^4 + 6.643432588 u^5 + O(u^6),$$

and

$$h(u) = 1 - u - 13.2294089033 u^4 + O(u^5).$$

Hence both metric functions reproduce the Schwarzschild form

$$f_{\text{Schw}} = h_{\text{Schw}} = 1 - u$$

through cubic order in the weak-field compactness variable. The strong-field P5 deformation begins only at the next displayed order. This provides a sharper statement than merely saying that the geometry has an acceptable asymptotic limit.

3.6 Horizonlessness and the regular core

The autonomous closure has

$$\Xi(u) \longrightarrow 1 \quad (u \rightarrow \infty).$$

Consequently,

$$f \longrightarrow \frac{1}{4}$$

rather than zero.

A direct global scan of the radial closure gives

$$h_{\min} \simeq 0.3268566652 > 0$$

at approximately

$$\Xi \simeq 0.764016, \quad u \simeq 0.6786844, \quad x \simeq 1.473439.$$

Thus the frozen P5 closure possesses no metric horizon of the form $\mathbf{f}=0$ or $\mathbf{h}=0$. In the strict causal terminology used in the compact-object literature it is therefore more accurately classified as a **regular horizonless ultracompact geometry** than as a regular black hole.

The central asymptotics follow from

$$\mathcal{P}_5(1) = 112.9263595.$$

Writing $\epsilon = 1 - \Xi$, the leading equation is

$$\frac{d\epsilon}{du} \simeq -\mathcal{P}_5(1)\epsilon^{3/2},$$

so that

$$1 - \Xi \sim \frac{4}{\mathcal{P}_5(1)^2 u^2} \simeq \frac{3.13667365 \times 10^{-4}}{u^2}.$$

It follows that

$$f = \frac{1}{4} + \frac{7.84168413 \times 10^{-5}}{u^2} + O(u^{-4}),$$

and

$$h = 1 - \frac{2.18115983 \times 10^{-3}}{u^2} + O(u^{-4}).$$

Since $\mathbf{u} = \mathbf{r}_s / \mathbf{r}$, both corrections are even and quadratic in $\mathbf{r}$ near the center. The zeroth-order metric is

$$ds^2 \simeq -\frac{1}{4}dt^2 + dr^2 + r^2 d\Omega^2,$$

which becomes locally Minkowskian at leading order after the constant time rescaling $\mathbf{T} = \mathbf{t}/2$; the $O(r^2)$ corrections give finite rather than divergent curvature. This statement should not be confused with an asymptotic Minkowski normalization of the original time coordinate at the center.

3.7 Background trajectory in invariant space

Because $\Xi(\mathbf{r})$ is monotonic on all reconstruction intervals, it provides a convenient coordinate along the one-dimensional background trajectory

$$r \mapsto (\phi(r), X(r), F(r), Y(r)), \quad \phi = \Xi.$$

For the static radial scalar,

$$X(r) = -\frac{1}{2}h(r)\phi'(r)^2.$$

For the electric background

$$A_\mu = (A_0(r), 0, 0, 0),$$

we use

$$F_{\text{bg}} = \frac{hA_0'^2}{2f},$$

and, in the convention adopted by the numerical reconstruction,

$$Y_{\text{bg}} = 4XF_{\text{bg}}.$$

The fact that the background occupies only a one-dimensional curve inside the multidimensional invariant space is the origin of the transverse action-jet freedom used later to stabilize perturbations without deforming the frozen geometry.

3.8 Reproducibility status

The P5 susceptibility, temporal closure, and radial closure have been independently regressed against the frozen source profile. The resulting geometry can now be regenerated from the equations in this section and the boundary condition $\Xi(0)=0$.

The historical SSZ 1.8r_s–2.2r_s Hermite bridge remains important as provenance and as a variational template for smooth handovers, but it is **not** inserted into the present strong-field P5 geometry as a fundamental field equation. The distinction between the historical bridge and the current P5 closure is maintained throughout the rest of the paper.

4. Null Geometry of the P5 Strong-Field Region

4.1 Circular-null condition

For equatorial null geodesics,

$$\theta = \frac{\pi}{2},$$

the Killing symmetries imply conserved quantities

$$E = f\dot{t}, \quad L_z = r^2\dot{\varphi}.$$

The null condition,

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0,$$

then yields

$$-\frac{E^2}{f} + \frac{\dot{r}^2}{h} + \frac{L_z^2}{r^2} = 0.$$

Equivalently,

$$\dot{r}^2 = h \left(\frac{E^2}{f} - \frac{L_z^2}{r^2} \right).$$

Introducing the impact parameter

$$b = \frac{L_z}{E},$$

circular null trajectories satisfy

$$\dot{r} = 0$$

and

$$\frac{d}{dr}\dot{r}^2 = 0.$$

The first condition implies

$$b^2 = \frac{r^2}{f(r)}.$$

The second implies

$$\frac{d}{dr} \left(\frac{r^2}{f} \right) = 0,$$

or

$$\boxed{rf' - 2f = 0.}$$

The P5 geometry possesses two strong-field roots of this expression.

4.2 Outer unstable light ring

The outer solution is

$$\boxed{x_{\text{out}} = 1.5}.$$

Its instability is determined by the sign of the second variation of the null effective potential.

This light ring was used as one of the principal geometrical calibration points of the P5 strong-field profile.

At this orbit the construction reproduces the desired Schwarzschild optical scale.

In the units $\mathbf{r_s=2GM}$,

$$r_{\text{out}} = 1.5r_s = 3GM,$$

coinciding with the Schwarzschild photon-sphere radius.

The corresponding critical impact parameter is therefore arranged to reproduce the Schwarzschild shadow scale.

The eikonal instability rate, characterized by the radial Lyapunov exponent of the orbit, is also matched at this outer ring.

The significance of the present dynamical reconstruction is that this geometrical agreement does not coincide with a degeneracy of the local field-theory kinetic matrix.

A full-SVT background pivot investigated during the construction has determinant

$$\det \mathcal{M} = f^{5/2} \sqrt{h} \phi'^2 r^8,$$

up to the normalization adopted for the particular local reconstruction variables.

Crucially, this expression contains no factor

$$r f' - 2f.$$

Hence the light-ring equation is not automatically a perturbative determinant zero.

Numerically the pivot remains finite at the outer orbit.

This already indicates that the optical critical point of null geodesics and the constraint structure of the covariant field theory are logically distinct.

4.3 Inner stable light ring

The second root occurs at

$$x_{\text{in}} \simeq 1.41616064$$

or

$$u_{\text{in}} \simeq 0.706135.$$

The second derivative of the null potential has the opposite sign from that of the outer orbit, making this solution stable against infinitesimal radial null displacements at the geodesic level.

The associated null well is shallow.

The numerically reconstructed relative depth is approximately

$$\Delta_{\text{well}} \simeq 4.177 \times 10^{-3}.$$

This shallow trapping region is one of the most interesting aspects of the geometry.

It also forces us to distinguish three different notions of stability.

First, there is **geodesic stability** of a circular null trajectory.

Second, there is **local field-theory stability**, meaning absence of ghosts and imaginary characteristic speeds.

Third, there is **global dynamical stability**, which concerns the long-time evolution of finite perturbations and includes possible effects of trapping and mode accumulation.

The inner light ring is stable in the first sense.

The genuine-SVT realization constructed below is healthy in the second sense.

The third question remains open.

5. Covariant Theory Architecture

5.1 Full Horndeski sector

We use the standard four-dimensional Horndeski action in the generalized-Galileon form

$$S_H = \int d^4x \sqrt{-g} \sum_{i=2}^5 \mathcal{L}_i,$$

with

$$X = -\frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi.$$

Our sign convention is

$$\boxed{\mathcal{L}_2 = G_2(\phi, X),}$$

$$\boxed{\mathcal{L}_3 = -G_3(\phi, X) \square \phi,}$$

$$\boxed{\mathcal{L}_4 = G_4(\phi, X) R + G_{4,X} [(\square \phi)^2 - \phi_{;\mu\nu} \phi^{;\mu\nu}],}$$

and

$$\boxed{\begin{aligned} \mathcal{L}_5 = & G_5(\phi, X) G_{\mu\nu} \phi^{;\mu\nu} \\ & - \frac{G_{5,X}}{6} \left[(\square \phi)^3 - 3 \square \phi \phi_{;\mu\nu} \phi^{;\mu\nu} + 2 \phi_{;\mu}^{;\nu} \phi_{;\nu}^{;\rho} \phi_{;\rho}^{;\mu} \right]. \end{aligned}}$$

Here

$$\phi_{;\mu\nu} \equiv \nabla_\mu \nabla_\nu \phi, \quad G_{i,X} = \frac{\partial G_i}{\partial X}.$$

These combinations are arranged so that the equations of motion remain second order despite the derivative interactions. The Einstein-Hilbert term is included as the special choice $G_4 = M_{\text{Pl}}^2/2$.

The P5 construction does not require all four functions to be active with equal importance at every radius. The outer carrier uses a restricted luminal sector with

$$G_4 = G_4(\phi), \quad G_5 = 0,$$

and a finite cubic interaction

$$G_3(\phi, X).$$

The regular center instead uses genuine quartic Horndeski jets with nonzero $G_{\{4,X\}}$ -type structure and, where convenient in the local jet representation, the equivalent $G_{\{5,\phi\}}$ control.

The full $\mathcal{L}_2$ – $\mathcal{L}_5$ definition is given here because later perturbative coefficients such as

$$\mathcal{F}, \quad \mathcal{G}, \quad \mathcal{H},$$

and the common even-parity coefficients a_i, b_i, c_i, d_i, e_i depend on these derivatives. Treating the action only schematically obscures which radial jets are independent and which are fixed by integrability.

5.2 Full gauge-invariant scalar–vector–tensor sector

We introduce a $U(1)$ gauge field

$$A_\mu \rightarrow A_\mu + \nabla_\mu \Lambda,$$

with

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu, \quad \tilde{F}^{\mu\nu} = \frac{1}{2} \mathcal{E}^{\mu\nu\alpha\beta} F_{\alpha\beta}.$$

The standard scalar-vector invariants are

$$X = -\frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi,$$

$$F = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

$$\tilde{F} = -\frac{1}{4} F_{\mu\nu} \tilde{F}^{\mu\nu},$$

and

$$Y = \nabla_\mu \phi \nabla_\nu \phi F^{\mu\alpha} F^\nu{}_\alpha.$$

A convenient representation of the genuine second-order gauge-invariant SVT interactions is

$$\boxed{\mathcal{L}_{\text{SVT}}^{(2)} = f_2(\phi, X, F, \tilde{F}, Y),}$$

$$\mathcal{L}_{\text{SVT}}^{(3)} = \mathcal{M}_3^{\mu\nu} \nabla_\mu \nabla_\nu \phi,$$

with

$$\mathcal{M}_3^{\mu\nu} = \left[f_3(\phi, X) g_{\rho\sigma} + \tilde{f}_3(\phi, X) \nabla_\rho \phi \nabla_\sigma \phi \right] \tilde{F}^{\mu\rho} \tilde{F}^{\nu\sigma},$$

and

$$\mathcal{L}_{\text{SVT}}^{(4)} = \mathcal{M}_4^{\mu\nu\alpha\beta} \nabla_\mu \nabla_\alpha \phi \nabla_\nu \nabla_\beta \phi + f_4(\phi, X) L^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta},$$

where

$$\mathcal{M}_4^{\mu\nu\alpha\beta} = \left[\frac{1}{2} f_{4,X}(\phi, X) + \tilde{f}_4(\phi) \right] \tilde{F}^{\mu\nu} \tilde{F}^{\alpha\beta},$$

and the double dual Riemann tensor is

$$L^{\mu\nu\alpha\beta} = \frac{1}{4} \mathcal{E}^{\mu\nu\rho\sigma} \mathcal{E}^{\alpha\beta\gamma\delta} R_{\rho\sigma\gamma\delta}.$$

These interactions propagate one scalar, two vector, and two tensor degrees of freedom on the generic background, for a total of five physical modes.

For the static P5 reconstruction we use

$$A_\mu = (A_0(r), 0, 0, 0),$$

so the parity-odd invariant $\tilde{\mathbf{F}}$ vanishes on the background. Its higher derivatives can nevertheless influence odd perturbations and therefore provide background-null control directions.

The central constructive representative is defined along the background trajectory by the jet choices

$$f_{2,F} = 1, \quad f_{2,Y} = 0,$$

$$\tilde{f}_4 = 0,$$

$$f_{4,X} = N_4, \quad f_{4,XX} = 0,$$

while

$$f_{3,X}$$

is not prescribed as a free pointwise stabilizer. It is solved from the scalar background equation, as described in Sec. 7.

5.3 One common action and the no-double-counting convention

The most useful architecture for the final P5 completion is

$$S = \int d^4x \sqrt{-g} \left[\sum_{i=2}^5 \mathcal{L}_i^{\text{H}} + \Delta \mathcal{L}_{\text{SVT}}^{\text{genuine}} \right].$$

The symbol Δ is important. The general SVT $\mathbf{f}_2$ sector can itself contain pure scalar and Maxwell pieces already represented in the Horndeski/Maxwell baseline. In the hybrid bookkeeping these shared pieces are counted exactly once.

Equivalently, at the level of any unreduced quadratic-action coefficient,

$$C_{\text{HSVT}} = C_{\text{MH}} + (C_{\text{SVT}} - C_{\text{shared}}).$$

This combination must be performed **before** constraint elimination.

The P5 construction therefore distinguishes three kinds of action data:

1. **background-constrained jets**, fixed by the metric, scalar, vector and background equations;
2. **background-null transverse jets**, unconstrained by the one-dimensional background curve but visible to perturbations;
3. **handover jets**, generated when different healthy local representatives are embedded into one scalar-coordinate partition and differentiated.

This classification is the basis of the same-action closure proved in Sections 16–21.

5.4 Static field equations and integrability hierarchy

On the static spherical ansatz the independent background equations may be organized as

$$\mathcal{E}_{00} = 0, \quad \mathcal{E}_{11} = 0, \quad \mathcal{E}_{\phi} = 0, \quad \mathcal{E}_A = 0.$$

The reconstruction does not use only the metric equations. In the strong-field lobe the metric equations determine the required on-shell values of $\mathbf{f}_2$ and $\mathbf{f}_{\{2,\mathbf{X}\}}$, the vector equation fixes the electric branch/current condition, and the scalar equation is retained as a nontrivial first-order integrability equation for $\mathbf{f}_{\{3,\mathbf{X}\}}$.

This ordering prevents a common inverse-construction error: a set of perturbatively convenient pointwise jets is not accepted unless it can be lifted to a local action section that also satisfies the background scalar and vector equations.

The resulting theory is therefore reconstructed around the frozen P5 geometry, but it is not merely fitted to a perturbation matrix.

6. Exterior Horndeski Reconstruction

6.1 Motivation

The genuine-SVT vector background becomes weak near its natural onset.

Trying to force the complete strong-field stabilization problem onto pure Einstein-plus-SVT precisely where

$$A'_0 \rightarrow 0$$

can drive required interaction derivatives to unnaturally large values.

The correct solution is not to insist that one sector carry every radius.

Instead, the exterior scalar–tensor sector carries the solution through the vector onset.

The genuine-SVT interactions then become increasingly important inward.

6.2 Correct general- $f \neq h$ no-ghost relation

For the restricted luminal exterior carrier,

$$G_4 = G_4(\phi), \quad G_5 = 0,$$

the scalar kinetic combination can be written in terms of an auxiliary function $Q(r)$.

A previously used simplified expression was valid only in the special case

$$f = h.$$

The frozen P5 background does not satisfy this equality throughout the relevant exterior region.

Using the general Maxwell–Horndeski relation, define

$$y(r) = \frac{\sqrt{f/h}}{Q}.$$

The scalar no-ghost reconstruction becomes a first-order equation

$$\boxed{y' = \frac{1}{2} \sqrt{\frac{f}{h}} (K + H),}$$

where

$$H = 2G_4$$

for the restricted carrier.

Imposing the regular endpoint conditions fixes a positive constant scalar kinetic margin of approximately

$$K_{\text{ext}} \simeq 9.7423 \times 10^{-3}.$$

This correction is important conceptually.

The exterior Horndeski witness survives the use of the full $\mathbf{f} \neq \mathbf{h}$ geometry.

6.3 Continuation through the strong-field region

The corrected Horndeski carrier can be continued inward while remaining finite and retaining positive scalar kinetic energy.

In particular, a smooth $H(\mathbf{r})$ profile can be constructed between the outer onset and the quartic-core boundary such that the associated cubic interaction $G_{\{3,X\}}$ remains finite.

This continuation passes through

$$x = 1.5$$

and

$$x \simeq 1.41616064.$$

Thus no singular Horndeski control pivot occurs at either light ring.

A direct full-Horndeski principal-control rank test was carried out using a nonredundant set of targets,

$$(F, H, a_1, c_4, c_2),$$

and controls drawn from

$$G_{4,X}, \quad G_{4,XX}, \quad G_{4,\phi X}, \quad G_{5,\phi}, \quad G_{5,\phi X}, \quad G_{5,X}, \quad G_{5,XX}, \quad G_{2,XX}.$$

The control map retains full rank across the onset-to-core interval.

The apparent degeneracy that appears if one attempts to control redundant tensor combinations precisely at

$$rf' - 2f = 0$$

is therefore a target-space redundancy associated with the photon-sphere condition, not a genuine disappearance of the Horndeski action controls.

This will become important when discussing the hybrid light-ring transition.

7. Genuine-SVT Strong-Field Lobe

7.1 Domain

The background-integrable strong-field SVT representative is constructed on

$$\boxed{0.61 \leq u \leq 0.715.}$$

Since

$$x = \frac{1}{u},$$

this corresponds to

$$\boxed{1.3986014 \lesssim x \lesssim 1.6393443.}$$

Both light rings lie strictly inside this interval.

The outer light ring has

$$u = \frac{2}{3} \simeq 0.666667.$$

The inner stable ring has

$$u \simeq 0.706135.$$

The inner end of the SVT lobe extends beyond the start of the regular Horndeski core,

$$u_{\text{core}} = 0.71,$$

giving a genuine radial overlap.

7.2 Background equations

The static field equations may be written schematically as

$$\mathcal{E}_{00} = 0,$$

$$\mathcal{E}_{11} = 0,$$

$$\mathcal{E}_{\phi} = 0,$$

$$\mathcal{E}_A = 0.$$

For the zero-net-charge branch used here, the vector equation can be represented through the corresponding conserved current condition.

The two independent metric equations are used to determine the on-shell values of

$$f_2$$

and

$$f_{2,X}.$$

The scalar equation is not discarded.

It is used to solve for the previously free quantity

$$q(r) \equiv f_{3,X}|_{\text{bg}}.$$

After substituting all other chosen jets and implementing the background-chain-rule relations, the scalar field equation becomes a linear first-order ordinary differential equation for $q(r)$.

This is one of the most important constructive steps in the calculation.

A pointwise assignment of $f_{3,X}$ could always be used to alter a perturbation coefficient.

It would not demonstrate that the assigned quantity belongs to an action whose scalar equation is solved by the P5 profile.

Solving the ODE does.

7.3 Integrability result

A minimax-boundary solution was obtained throughout the complete interval

$$0.61 \leq u \leq 0.715.$$

The reconstructed derivative remains finite, with

$$\max |f_{3,X}| \simeq 44.73.$$

The direct residual of the scalar equation is approximately

$$\max |\mathcal{E}_\phi| \lesssim 2 \times 10^{-12}.$$

Restricting the domain to

$$u \leq 0.71$$

reduces the maximum required magnitude to approximately

The extension to $u=0.715$ is nevertheless useful because it creates a finite overlap with the inner Horndeski core.

The strong-field SVT construction should therefore be regarded as an explicit **on-shell action-jet section**, rather than merely a perturbative coefficient fit.

8. Tubular Reconstruction and Background-Null Controls

8.1 Why higher action derivatives are not fixed by the background

The static background traces only a one-dimensional trajectory through the multidimensional invariant space

$$(\phi, X, F, Y).$$

Knowledge of the value and first directional derivative of an action function along this curve does not determine all derivatives transverse to the curve.

For example, suppose

$$f_2 = f_2(\phi, X, F, Y).$$

The background equations constrain combinations involving quantities such as

$$f_{2,X}, \quad f_{2,F}, \quad f_{2,Y},$$

evaluated on the trajectory.

They do not uniquely determine the entire Hessian

$$\nabla^2 f_2 = \begin{pmatrix} f_{2,XX} & f_{2,XF} & f_{2,XY} \\ f_{2,XF} & f_{2,FF} & f_{2,FY} \\ f_{2,XY} & f_{2,FY} & f_{2,YY} \end{pmatrix}.$$

Those transverse second derivatives enter the perturbation operator.

This creates a natural set of controls that can change stability while leaving the background unchanged.

8.2 Hessian-control basis

We parameterize the independent background-null Hessian directions by

$$H = (H_{XX}, H_{XF}, H_{XY}, H_{FF}, H_{FY}, H_{YY}).$$

For the central lobe, the perturbative quantities v_1 , v_4 , and c_2 depend linearly on these controls. The relevant linear maps may be represented by the rows

$$L_{v_1} = (0, 0, 0, 1, -4\kappa, 4\kappa^2),$$

where

$$\kappa = h\phi'^2,$$

and

$$L_{v_4} = (0, -1, 2\kappa, 0, -4F_{\text{bg}}, 8F_{\text{bg}}\kappa).$$

For the scalar radial coefficient,

$$L_{c_2} = \left(\frac{\kappa}{2}, -F_{\text{bg}}, 6F_{\text{bg}}\kappa, 0, -4F_{\text{bg}}^2, 16F_{\text{bg}}^2\kappa\right).$$

The corresponding radial prefactors are finite throughout the lobe.

A minimum-norm Hessian solution therefore exists for the chosen perturbative targets.

The largest Euclidean Hessian norm required in the complete interval is approximately

$$\|H\|_{\text{max}} \simeq 4.27 \times 10^2.$$

No loss of control rank occurs in the light-ring region.

9. Odd-Parity Stability

9.0 Harmonic decomposition and the complete odd reduction chain

Before quoting the numerical inequalities, it is useful to make the perturbation reduction explicit.

For each ℓ, m , define the axial vector harmonic

$$S_a^{\ell m} = E_a^{b} \nabla_b Y_{\ell m}.$$

Up to conventional normalization of the tensor harmonic, the odd metric perturbations are represented by

$$\delta g_{ta} = Q(t, r) S_a^{\ell m}, \quad \delta g_{ra} = W(t, r) S_a^{\ell m},$$

and an angular tensor amplitude $U(\mathbf{t}, \mathbf{r})$. The odd vector perturbation is

$$\delta A_a = \delta A(t, r) S_a^{\ell m}.$$

There is no odd scalar perturbation. For $\ell \geq 2$, the Regge–Wheeler-type gauge

$$U = 0$$

removes the remaining odd coordinate gauge freedom.

The quadratic odd action has the generic reduced form

$$S_{\text{odd}}^{(2)} = \frac{1}{2} \int dt dr [\dot{\mathbf{q}}_o^T K_o \dot{\mathbf{q}}_o + \dot{\mathbf{q}}_o^T R_o \mathbf{q}'_o - \mathbf{q}'_o{}^T G_o \mathbf{q}'_o - \mathbf{q}_o^T M_o \mathbf{q}_o],$$

with two physical fields after the nondynamical metric amplitude has been removed,

$$\mathbf{q}_o = (q_T, q_V)^T.$$

In the Maxwell–Horndeski limit this reduction can be written using an auxiliary gravitational field χ_o so that the dynamical pair is $(\chi_o, \delta A)$. The pure-limit action provides an important regression target: the tensor and vector no-ghost conditions reduce to positivity of the corresponding kinetic entries, and the eikonal characteristic equation is

$$\det(\omega^2 K_o - \omega k R_o - k^2 G_o) = 0.$$

The genuine-SVT theory changes the coefficients and allows additional mixing, but not the physical mode count. In the Zhang–Kase coefficient basis used by the present numerical reconstruction, the required local inequalities reduce to the α_i conditions stated below. The entire odd scan is therefore a test of the **two-field quadratic operator**, not a collection of unrelated scalar inequalities.

The parity-odd background-null derivative $f_{\{2, \tilde{F} \tilde{F}\}}$ is kept explicit because $\tilde{F}=0$ on the electric background while its second derivative enters the odd angular principal symbol. This is a concrete example of a tubular action direction that is invisible to the background but physical at quadratic order.

9.1 Structure of the sector

Odd-parity perturbations contain the tensor-like gravitational mode together with the transverse vector perturbation.

The local health conditions are conveniently written in terms of the background coefficients

$$\alpha_1, \dots, \alpha_9.$$

The exact definitions are those of the general gauge-invariant SVT perturbation formalism.

The central construction produces

$$\alpha_4 > 0,$$

$$\alpha_7 > 0,$$

and

$$\alpha_6 < 0.$$

The vector kinetic combination

$$C_{\text{odd}} \equiv \alpha_2^2 - \alpha_1 \alpha_5$$

must also remain positive.

Numerically,

$$\boxed{\min C_{\text{odd}} \simeq 0.76072.}$$

The minimum values obtained over the lobe include approximately

$$\boxed{\min \alpha_4 \simeq 9.996 \times 10^{-3},}$$

and

$$\boxed{\min \alpha_7 \simeq 2.0015.}$$

Thus neither the tensor nor vector odd-parity kinetic block approaches zero.

9.2 Radial odd speeds

The tensor radial squared propagation speed is

$$c_{r,T}^2 = -\frac{\alpha_6}{fh \alpha_7}.$$

Throughout the complete strong-field lobe,

$$\boxed{\min c_{r,T}^2 \simeq 1.00507.}$$

The odd-vector radial speed takes the form

$$c_{r,V}^2 = \frac{\alpha_2^2 - \alpha_1 \alpha_5}{fh \alpha_1 \alpha_4}.$$

The minimum obtained numerically is

$$\boxed{\min c_{r,V}^2 \simeq 9.45887.}$$

Both modes therefore remain radially hyperbolic.

The large value of the vector speed is noted but not interpreted here as an instability.

The mathematical condition relevant to the local Laplacian analysis is

$$c_r^2 > 0.$$

Questions concerning superluminal effective characteristics are postponed to a dedicated causality/EFT analysis.

9.3 Odd angular sector

The tensor angular coefficient remains positive over the same interval.

The vector angular mode contains an additional action derivative that can be adjusted along a background-null direction.

A particularly useful control is the second derivative with respect to the parity-odd electromagnetic invariant,

$$f_{2,\tilde{F}\tilde{F}},$$

which does not affect the static electric background or the even linear sector.

This permits the odd angular coefficient α_9 to be tuned independently.

Choosing the corresponding background-null jet so that

$$\alpha_9 = -1$$

provides a strictly positive lower bound for the remaining vector angular numerator.

The resulting margin is bounded below by a term proportional to

$$4\alpha_7,$$

whose minimum is approximately

$$8.006.$$

Thus the odd angular sector can be completed without disturbing the already-fixed background or even-sector construction.

10. Even-Parity Stability

10.0 Even harmonic content, gauge fixing, and the unreduced action

For $\ell \geq 2$, the even metric sector contains the scalar amplitudes

$$H_0, \quad H_1, \quad H_2, \quad h_0, \quad h_1, \quad K, \quad G,$$

while

$$\phi = \bar{\phi}(r) + \delta\phi(t, r)Y_{\ell m}$$

and the vector perturbation is decomposed as

$$\delta A_t = \delta A_0 Y_{\ell m}, \quad \delta A_r = \delta A_1 Y_{\ell m}, \quad \delta A_a = \delta A_2 \nabla_a Y_{\ell m}.$$

We use the uniform-curvature-type coordinate gauge

$$h_0 = 0, \quad G = 0, \quad K = 0,$$

and the U(1) gauge

$$\delta A_2 = 0.$$

The remaining unreduced variables are therefore

$$\boxed{(H_0, H_1, H_2, h_1, \delta\phi, \delta A_0, \delta A_1, V)},$$

where V is the auxiliary vector-flux variable introduced below.

The Maxwell–Horndeski pure limit provides a useful transparent template for the common action. Its H_0 constraint sector contains

$$\begin{aligned} \mathcal{L}_u \supset H_0 [a_1 \delta\phi'' + a_2 \delta\phi' + a_3 H_2' + L a_4 h_1' \\ + (a_5 + L a_6) \delta\phi + (a_7 + L a_8) H_2 + L a_9 h_1], \end{aligned}$$

with $L = \ell(\ell+1)$. The electric perturbation sector has the schematic exact structure

$$\begin{aligned} \mathcal{L}_A = v_1 (\delta A_0' - \delta A_1')^2 \\ + (\delta A_0' - \delta A_1') (v_2 H_0 + v_3 H_2 + v_4 \delta\phi' + v_5 \delta\phi) + v_6 H_0^2 + \dots \end{aligned}$$

Introducing V rewrites the square as

$$\begin{aligned} \mathcal{L}_A = v_1 \left[2V \left(\delta A_0' - \delta A_1' + \frac{v_2 H_0 + v_3 H_2 + v_4 \delta\phi' + v_5 \delta\phi}{2v_1} \right) - V^2 \right] \\ - \frac{(v_2 H_0 + v_3 H_2 + v_4 \delta\phi' + v_5 \delta\phi)^2}{4v_1} + \dots \end{aligned}$$

Variation with respect to $\mathbf{V}$ returns

$$V = \delta A'_0 - \delta A_1 + \frac{v_2 H_0 + v_3 H_2 + v_4 \delta \phi' + v_5 \delta \phi}{2v_1}.$$

The genuine-SVT theory extends the vector coefficient dictionary to $\mathbf{v}_1, \dots, \mathbf{v}_{13}$ and modifies the scalar-metric coefficients, but the same unreduced architecture survives. The role of H_0 as a Lagrange multiplier and the need to eliminate common constraints **after** assembling all Horndeski and SVT contributions are therefore structural rather than model-specific.

After all nondynamical fields are removed, the three physical even modes can be represented by

$$\mathbf{q}_e = (\psi, \delta\phi, V)^T,$$

with quadratic principal action

$$S_{\text{even}}^{(2)} = \frac{1}{2} \int dt dr [\dot{\mathbf{q}}_e^T K_e \dot{\mathbf{q}}_e + \dot{\mathbf{q}}_e^T R_e \mathbf{q}'_e - \mathbf{q}'_e^T G_e \mathbf{q}'_e - \mathbf{q}_e^T M_e \mathbf{q}_e + \dots].$$

Thus the final no-ghost condition is positivity of the physical kinetic matrix K_e , while radial characteristics are roots of

$$\det(\omega^2 K_e - \omega k R_e - k^2 G_e) = 0.$$

The large- L angular symbol supplies the independent angular characteristic test. Sections 10–12 evaluate these conditions for the pure genuine-SVT P5 lobe; Sections 18–21 construct the common hybrid operator needed for the same-action interpolation.

10.1 Why this is the decisive sector

The even-parity system is substantially more complicated than the odd sector because the scalar perturbation couples to the even gravitational and vector modes.

For $\ell \geq 2$, the complete theory contains three physical even-parity propagating degrees of freedom.

Their no-ghost and gradient conditions cannot generally be inferred from the separate scalar, vector, and tensor subsectors.

The coupled kinetic and angular matrices must be analyzed.

This makes the successful even-sector reconstruction the strongest local result of the present work.

10.2 Exact $\mathbf{P}_1$ definition

A crucial consistency correction arose during the calculation because the typesetting of the published expression had initially been parsed incorrectly.

With the adopted conventions,

$$P_1 = \frac{a_4}{f} \frac{d}{dr} \left(r \sqrt{\frac{f}{h}} \right).$$

For the Einstein-plus-genuine-SVT central representative,

$$a_4 = \frac{\sqrt{fh}}{2}.$$

There is no additional factor of r multiplying a_4 .

This correction is important because an earlier provisional implementation had used

$$a_4^{\text{old}} = \frac{r\sqrt{fh}}{2},$$

which is incorrect.

All final values quoted in this manuscript use the corrected expression.

10.3 Exact even no-ghost combination

The relevant kinetic combination is

$$\mathcal{K} = 2 \left(P_1 + \frac{rA'_0 v_6}{2\sqrt{fh}} \right) - 4\sqrt{fh} \alpha_7 r^2.$$

The Appendix-A coefficients provide an independent internal identity,

$$e_1 = \frac{\mathcal{K}}{\sqrt{fh} \phi'^2}.$$

This identity constitutes an unusually strong regression test because e_1 and $\mathcal{K}$ are assembled from different combinations of the unreduced quadratic-action coefficients.

On the complete background-integrable lobe we find

$$\max \left| \mathcal{K} - e_1 \sqrt{fh} \phi'^2 \right| \simeq 2.18 \times 10^{-10}.$$

The identity therefore closes numerically.

The kinetic combination itself satisfies

$$10.5415 \lesssim \mathcal{K} \lesssim 137.879.$$

In particular,

$$\mathcal{K} > 0$$

throughout the entire interval.

At the outer light ring,

$$x = 1.5,$$

we obtain approximately

$$\mathcal{K}_{\text{out}} \simeq 23.3092.$$

At the inner stable light ring,

$$x \simeq 1.41616064,$$

we obtain

$$\mathcal{K}_{\text{in}} \simeq 71.1498.$$

Thus neither light ring comes remotely close to the even-sector kinetic boundary.

11. Radial Even-Parity Characteristics

The three radial propagation modes correspond locally to tensor, scalar, and vector characteristic branches.

For the constructed strong-field lobe the tensor mode satisfies

$$\min c_{r,T}^2 \simeq 1.005069.$$

The corrected scalar mode is no longer identically unity.

The original Hessian target had been selected using the provisional P_1 parse, which artificially tuned

$$c_{r,S}^2 = 1$$

to floating-point precision.

After inserting the exact P_1 and $\mathcal{K}$, while retaining the **same already-integrable action representative**, the scalar mode becomes

$$0.903745 \lesssim c_{r,S}^2 \lesssim 1.061333.$$

The correction therefore does not destroy stability.

It simply reveals that the original unit-speed target was slightly mistuned.

At the outer light ring,

$$c_{r,S}^2(x = 1.5) \simeq 1.000018.$$

At the inner stable light ring,

$$c_{r,S}^2(x \simeq 1.41616064) \simeq 0.91652.$$

The vector radial branch satisfies

$$\min c_{r,V}^2 \simeq 9.45887.$$

Hence

$$c_{r,i}^2 > 0 \quad \text{for all three even radial branches over the complete lobe.}$$

There is therefore no radial Laplacian instability in the constructed strong-field SVT region.

12. Coupled Even Angular Sector

12.1 Need for the full coupled test

Positivity of radial speeds is not sufficient.

In spherical backgrounds the high- ℓ angular principal symbol produces an independent set of characteristic conditions.

Moreover, the scalar and vector angular branches are coupled.

The full angular system was therefore reconstructed using the Appendix-A coefficients

$$a_i, c_i, d_i, e_i, v_i,$$

including the nontrivial quantities

$$v_{13}, \quad d_3, \quad e_4,$$

and the associated matrix combinations conventionally denoted

$$m_1, \dots, m_5$$

and

$$M_{ij}^{(0)}, \quad M_{ij}^{(1)}.$$

A particularly important implementation detail concerns

$$\partial_\phi v_6.$$

It must be evaluated as a true partial action derivative, not as the total background derivative

$$\frac{v'_6}{\phi'}.$$

The distinction matters because the action depends on several independent invariants whose background values all vary with $\mathbf{r}$.

The required partial derivative is reconstructed from the explicit action jets.

12.2 Tubular mixed derivatives

For the chosen background values

$$f_{2,F} = 1, \quad f_{2,Y} = 0,$$

the tubular construction yields mixed derivatives such as

$$f_{2,\phi F} = -\left(H_{XF}X_\phi + H_{FF}F_\phi + H_{FY}Y_\phi\right),$$

and

$$f_{2,\phi Y} = -\left(H_{XY}X_\phi + H_{FY}F_\phi + H_{YY}Y_\phi\right).$$

These terms enter the full angular coefficients.

The use of partial rather than total derivatives is essential for maintaining compatibility with a genuine multivariable action.

12.3 Numerical coupled conditions

The complete angular inequalities remain strongly positive.

For the first coupled condition,

$$\boxed{\min C_{\Omega,1} \simeq 738.609.}$$

For the second,

$$\boxed{\min C_{\Omega,2} \simeq 6.0303 \times 10^4.}$$

The associated discriminant satisfies

$$\min \Delta_\Omega \simeq 2.9938 \times 10^5.$$

The two coupled squared angular characteristic speeds are therefore real and positive.

Numerically,

$$\min c_{\Omega,-}^2 \simeq 84.912,$$

and

$$\min c_{\Omega,+}^2 \simeq 643.142.$$

These values are large, but the important local result is

$$c_{\Omega,\pm}^2 > 0.$$

Thus the full coupled even angular sector exhibits no Laplacian instability anywhere in the complete strong-field lobe.

13. Summary of the Genuine-SVT Light-Ring Result

Combining the preceding sections, the central SVT construction satisfies simultaneously

$$\mathcal{E}_{00} = \mathcal{E}_{11} = \mathcal{E}_\phi = \mathcal{E}_A \simeq 0$$

at numerical reconstruction accuracy,

$$\mathcal{K} > 0,$$

$$\alpha_4 > 0,$$

$$\alpha_7 > 0,$$

$$C_{\text{odd}} > 0,$$

$$c_{r,T}^2 > 0,$$

$$c_{r,S}^2 > 0,$$

$$c_{r,V}^2 > 0,$$

and

$$c_{\Omega,i}^2 > 0$$

for the complete set of locally propagating high-frequency branches covered by the odd/even analysis.

The result holds over

$$1.39860 \lesssim x \lesssim 1.63934.$$

Both light rings lie strictly inside this interval.

Therefore,

Neither the unstable nor the stable P5 light ring forces a local ghost or Laplacian instability.

This is one of the central results of the paper.

14. Regular Quartic-Horndeski Core

14.1 Why a separate core carrier is used

The cubic G_3 scalar carrier that works efficiently in the exterior cannot be extended arbitrarily deeply into the center while retaining the desired integrated stability budget.

Dedicated endpoint-integral and handover scans show that the available scalar G_3 stability budget becomes negative for sufficiently deep interior starting points.

This does not indicate a failure of Horndeski theory.

It indicates that the wrong Horndeski operator is being asked to carry the center.

The regular core is instead represented by a quartic-Horndeski construction.

14.2 Core domain

The present core witness covers approximately

$$10^{-6} \lesssim x \leq 1.4084507042.$$

Its outer boundary corresponds to

$$u = 0.71.$$

The central SVT lobe extends to

$$u = 0.715,$$

or

$$x \simeq 1.3986014.$$

Consequently, the two constructions overlap on

$$\boxed{1.39860 \lesssim x \lesssim 1.40845.}$$

This is a genuine overlap, not merely contact at one radius.

14.3 Quartic local action

A useful local quartic representative has the schematic form

$$G_4(\phi, X) = g(\phi) + C(\phi)X,$$

supplemented, where required, by the corresponding $\mathbf{G}_5$ -derivative jet represented locally through a function

$$G_{5,\phi} = D(\phi).$$

The core principal targets are chosen so that the tensor kinetic functions remain positive and approach their regular central limits.

The numerical witness yields approximately

$$\boxed{\min \mathcal{F} = 1,}$$

and

$$\boxed{\min \mathcal{G} = \min \mathcal{H} \simeq 0.999381.}$$

The scalar kinetic target remains positive over the finite radial grid.

Its minimum becomes extremely small near the regular center because of radial-coordinate scaling,

$$K_{\text{scalar},\min} \simeq 2.5 \times 10^{-15},$$

but the proper central Einstein-frame kinetic coefficient has the finite positive limit

$$\boxed{K_{\text{center}}^{\text{EF}} \simeq 95.89.}$$

The vanishing coordinate-space quantity must therefore not be misidentified as a physical strong-coupling zero without accounting for the regular center limit.

15. The Inner Handover

Because the genuine-SVT lobe extends to

$$u = 0.715$$

while the regular quartic-Horndeski core reaches outward to

$$u = 0.71,$$

there is a finite radial overlap

$$\boxed{0.71 \leq u \leq 0.715}$$

or, equivalently,

$$\boxed{1.3986014 \lesssim x \lesssim 1.4084507.}$$

The scalar profile is strictly monotone across this interval. With the normalization $\mathbf{r}_s=1$, the reconstructed P5 profile gives

$$\Xi(0.71) \simeq 0.84835352, \quad \Xi(0.715) \simeq 0.86076347,$$

and

$$\min |\phi'| \simeq 1.24037, \quad \phi \equiv \Xi.$$

Thus ϕ itself is a valid local patch coordinate throughout the handover.

The action-level gluing is performed with a **flat $\mathbf{C}^\infty$ partition** rather than with a compactly extended quintic. Let

$$t(\phi) = \frac{\phi - \phi_a}{\phi_b - \phi_a}.$$

For $0 < t < 1$ define

$$\boxed{s_\infty(t) = \frac{e^{-1/t}}{e^{-1/t} + e^{-1/(1-t)}}}$$

and extend it by $s_\infty=0$ for $t \leq 0$ and $s_\infty=1$ for $t \geq 1$. Every derivative of s_∞ vanishes at both endpoints. Consequently all finite action jets return exactly to their pure-sector

values at the patch boundaries.

This stronger endpoint property matters here. The full even-parity reducer contains radial derivatives of coefficient combinations that themselves depend on second and higher action derivatives. A partition that is only C^2 is therefore sufficient as a geometric interpolation, but not the cleanest choice for a finite-order same-action jet construction. The C^∞ partition removes this bookkeeping ambiguity completely.

Both endpoint theories are locally healthy before gluing. The handover problem is therefore not one of curing an instability, but of constructing a single smooth covariant action section that preserves the fixed P5 background and the strict perturbative inequalities through the overlap.

16. The Outer Horndeski–SVT Handover

16.1 Overlap domain

The natural strong/SVT onset occurs at

$$u_{\text{on}} = 0.5515230871346237,$$

or

$$x_{\text{on}} \simeq 1.81316072.$$

The explicit genuine-SVT witness is already available from $u=0.61$ inward, while the exterior Horndeski carrier remains healthy beyond that point. We therefore use the finite outer overlap

$$0.5515230871346237 \leq u \leq 0.61.$$

On the frozen P5 profile,

$$\Xi_a \simeq 0.50993310, \quad \Xi_b \simeq 0.60601158,$$

with

$$\min f \simeq 0.387706, \quad \min h \simeq 0.426991,$$

and

$$\min |\phi'| \simeq 0.446924.$$

Thus no metric or scalar-coordinate degeneracy occurs in the outer handover band.

The physical architecture is consequently an extended overlap,

16.2 What the historical SSZ bridge proves

The SSZ/JIF bridge corpus contains a mathematically useful but carefully limited result. If the endpoint value, first derivative and second derivative are prescribed at both ends of a normalized interval, then the functional

$$J_3[s] = \int_0^1 [s'''(t)]^2 dt$$

has the unique minimizer given by the quintic Hermite interpolant. For flat endpoint two-jets,

$$s(0) = 0, \quad s(1) = 1, \quad s'(0) = s'(1) = s''(0) = s''(1) = 0,$$

this is

$$s_{H5}(t) = 10t^3 - 15t^4 + 6t^5,$$

with

$$J_3[s_{H5}] = 720.$$

The variational proof follows from writing $\mathbf{s}=\mathbf{H}_5+\mathbf{q}$ with homogeneous endpoint two-jets:

$$J_3[H_5 + q] - J_3[H_5] = \int_0^1 (q''')^2 dt \geq 0.$$

This result is retained in the present work as a **geometric and variational reference**. It is not used as a dynamical field equation. The older bridge audit correctly emphasizes that infinitely many dynamical flows can share the same static radial profile.

There is also an important regularity distinction. When $\mathbf{H}_5$ is extended by constants outside $[0, 1]$,

$$H_5'''(0) = H_5'''(1) = 60,$$

so the extended partition is $\mathcal{C}^2$, not $\mathcal{C}^\infty$. The full same-action construction below therefore uses the flat $\mathcal{C}^\infty$ weight $\mathbf{s}_\infty$; the quintic remains the unique minimum- J_3 member of the fixed-two-jet class.

16.3 Scalar-coordinate same-action gluing

Because $\phi = \chi$ is monotone in both overlap bands, the transition can be written in field space rather than by inserting the coordinate $\mathbf{r}$ explicitly into the fundamental action. For every independent action function, introduce background-adapted normal coordinates $\mathbf{n}^A$ transverse to the one-dimensional background curve in invariant space. A local function may be written as

$$F(\phi, n) = F_0(\phi) + A_A(\phi)n^A + \frac{1}{2}H_{AB}(\phi)n^An^B + \frac{1}{3!}T_{ABC}(\phi)n^An^Bn^C + \dots$$

Only a finite jet order is required by the quadratic perturbation problem. Multiplying the normal expansion by a compact tubular cutoff away from the background curve gives an ordinary smooth local action function rather than a collection of unrelated pointwise coefficients.

The two endpoint representatives are glued at the level of the coefficient functions $F_0, A_A, H_{AB}, \dots$ with $s_{\infty}(\phi)$. Background-constrained coefficients are then **re-solved** from

$$\mathcal{E}_{00} = 0, \quad \mathcal{E}_{11} = 0, \quad \mathcal{E}_{\phi} = 0, \quad \mathcal{E}_A = 0,$$

rather than blindly interpolated. Background-null transverse derivatives are retained as stability controls.

This construction has three useful properties:

1. it defines one smooth local covariant action on a tubular neighborhood of the P5 background curve;
2. it recovers the pure representatives with all finite jets at the patch boundaries;
3. it leaves precisely the transverse action derivatives that were already identified as background-null perturbative controls.

The handover is therefore an action-space continuation problem, not an interpolation of reduced kinetic matrices.

17. Why Separately Reduced Matrices Must Not Be Added

Suppose one computes a reduced Maxwell–Horndeski even action

$$S_{\text{red}}^{\text{MH}}$$

after eliminating its nondynamical fields.

Suppose independently one computes

$$S_{\text{red}}^{\text{SVT}}.$$

It may appear tempting to write

$$S_{\text{hybrid}}^{\text{naive}} = S_{\text{red}}^{\text{MH}} + S_{\text{red}}^{\text{SVT}} - S_{\text{red}}^{\text{common}}.$$

This is not generally correct.

The reason is that the nondynamical fields are solved from algebraic or differential constraints whose coefficients depend on the full action.

If a nondynamical variable N appears schematically as

$$L = AN^2 + NB_i\dot{q}_i + C_{ij}\dot{q}_i\dot{q}_j,$$

then its solution is

$$N = -\frac{B_i\dot{q}_i}{2A}.$$

Substitution produces

$$L_{\text{red}} = \left(C_{ij} - \frac{B_i B_j}{4A} \right) \dot{q}_i \dot{q}_j.$$

For two interactions,

$$A = A_1 + A_2, \quad B = B_1 + B_2,$$

and consequently

$$\frac{(B_1 + B_2)^2}{A_1 + A_2} \neq \frac{B_1^2}{A_1} + \frac{B_2^2}{A_2}$$

in general.

Therefore reduction and addition do not commute.

The hybrid theory must be assembled before eliminating the common constraints.

18. Common Unreduced Even-Parity Schema

The common even-parity quadratic action has been organized in terms of the perturbation fields

$$\boxed{(H_0, H_1, H_2, h_1, \delta\phi, \delta A_0, \delta A_1, V)},$$

where V is an auxiliary vector-flux variable.

The common coefficient schema contains

$$a_1, \dots, a_9,$$

$$b_1, \dots, b_5,$$

$$c_1, \dots, c_6,$$

$$d_1, \dots, d_4,$$

$$e_1, \dots, e_4,$$

and

$$v_1, \dots, v_{13}.$$

The Maxwell–Horndeski vector numbering can be embedded consistently into the 13-slot genuine-SVT convention.

The mapping is

$$v_{1\dots 5}^{\text{MH}} \rightarrow v_{1\dots 5},$$

$$v_6^{\text{MH}} \rightarrow v_7,$$

$$v_7^{\text{MH}} \rightarrow v_8,$$

$$v_8^{\text{MH}} \rightarrow v_9,$$

$$v_9^{\text{MH}} \rightarrow v_{10},$$

with

$$v_6 = v_{11} = v_{12} = v_{13} = 0$$

for the pure Maxwell–Horndeski sector.

This bookkeeping is essential because otherwise coefficients carrying different meanings can be accidentally combined under identical labels.

19. Generalized ψ Variable

19.1 Apparent mismatch between the two perturbation formalisms

The separately derived Maxwell–Horndeski and genuine-SVT reductions appear at first to use different physical combinations of the metric perturbations.

This might suggest that the two systems cannot be smoothly combined.

The apparent conflict disappears at the unreduced level.

The derivative part of the common H_0 constraint contains

$$a_1 \delta \phi'' + a_3 H_2' + L a_4 h_1',$$

where

$$L = \ell(\ell + 1).$$

For nonzero a_3 ,

$$a_1 \delta \phi'' + a_3 H_2' + L a_4 h_1' = a_3 \frac{d}{dr} \left[H_2 + \frac{L a_4}{a_3} h_1 + \frac{a_1}{a_3} \delta \phi' \right]$$

at principal derivative order, with coefficient-derivative terms retained separately in the full jet-aware reduction.

This motivates the common definition

$$\psi = H_2 + \frac{L a_4}{a_3} h_1 + \frac{a_1}{a_3} \delta \phi'.$$

19.2 Pure genuine-SVT limit

For the Einstein-plus-genuine-SVT central representative,

$$a_1 = 0,$$

and

$$a_3 = -r a_4.$$

Therefore,

$$\frac{L a_4}{a_3} = -\frac{L}{r}.$$

Hence

$$\psi_{\text{SVT}} = H_2 - \frac{L}{r} h_1.$$

The numerical reconstruction verifies

$$\frac{La_4}{a_3} + \frac{L}{r} \simeq 0$$

to approximately 10^{-15} – 10^{-13} , depending on L .

The two published reduced-variable choices are therefore not incompatible.

They are limits of the same common unreduced constraint structure.

20. Jet-Aware Constraint Elimination

20.1 Why frozen-coefficient Fourier reduction fails

An early common-kernel implementation froze all radial coefficients at a point, replaced

$$\partial_t \rightarrow -i\omega, \quad \partial_r \rightarrow ik,$$

and then eliminated the nondynamical fields.

Although this procedure is often adequate for unconstrained systems at leading principal order, it fails here.

The published reduced kinetic matrices contain radial derivatives such as

$$a'_4,$$

and derivatives of compound constraint quantities such as

$$K'_2.$$

These terms are produced **during constraint elimination**.

Freezing the coefficient functions beforehand erases them.

Consequently,

$$\boxed{\text{freeze} \rightarrow \text{Fourier transform} \rightarrow \text{eliminate}}$$

does not reproduce the physical reduced principal operator.

The correct ordering is

$$\boxed{\text{combine radial action profiles} \rightarrow \text{retain coefficient jets} \rightarrow \text{eliminate common constraints} \rightarrow \text{take principal/eikonal limit}}$$

This ordering is now adopted throughout the hybrid construction.

20.2 First common elimination

Define

$$p(r) = \frac{a_1}{a_3},$$

and

$$q(r) = \frac{La_4}{a_3}.$$

Then

$$\psi = H_2 + qh_1 + p\delta\phi'.$$

Equivalently,

$$H_2 = \psi - qh_1 - p\delta\phi'.$$

Substitution into the complete H_0 constraint causes the coefficients of

$$\delta\phi''$$

and

$$h_1'$$

to cancel identically.

The remaining equation takes the form

$$0 = a_3\psi' + B_{\text{eff}}\delta\phi' + C_\phi\delta\phi + C_{H_2}\psi + D_{h_1}h_1 + v_2V.$$

Here

$$B = a_2 - \frac{v_2v_4}{2v_1},$$

$$C_\phi = a_5 + La_6 - \frac{v_2v_5}{2v_1},$$

$$C_{H_2} = a_7 + La_8 - \frac{v_2v_3}{2v_1},$$

and

$$C_{h_1} = L \left(a_9 - \frac{v_2v_6}{2v_1} \right).$$

After accounting for the radial derivatives of p and q ,

$$B_{\text{eff}} = B - a_3 p' - C_{H_2} p,$$

and

$$D_{h_1} = C_{h_1} - a_3 q' - C_{H_2} q.$$

Thus

$$h_1 = -\frac{a_3 \psi' + B_{\text{eff}} \delta \phi' + C_\phi \delta \phi + C_{H_2} \psi + v_2 V}{D_{h_1}}.$$

This is the first exact common Horndeski-SVT constraint-elimination step.

20.3 Pivot test

The denominator

$$D_{h_1}$$

could in principle vanish and produce a strong-coupling or constraint singularity.

We therefore evaluate it throughout the complete background-integrable SVT lobe.

For the lowest relevant gravitational multipole,

$$\ell = 2, \quad L = 6,$$

the minimum magnitude is

$$\min |D_{h_1}| \simeq 11.99.$$

No sign change occurs.

The test was extended through a wide range of angular eigenvalues up to approximately

$$L = 1000.$$

No zero of the pivot was found.

The absolute pivot grows rather than decreases with increasing L .

Consequently, the first common jet-aware elimination exhibits no constraint singularity in the light-ring region.

20.4 Exact elimination of the remaining SVT constraints

The remaining algebraic constraints can be written explicitly using the unreduced even-parity action of the general gauge-invariant SVT theory.

After the H_0 constraint has been used to express h_1 , define

$$J_1 = b_3 \dot{\delta\phi} + b_4 \dot{H}_2 + L b_5 \dot{h}_1.$$

The H_1 and δA_1 equations form the two-dimensional system

$$\begin{pmatrix} 2b_1 & v_{11} \\ v_{11} & 2v_{10} \end{pmatrix} \begin{pmatrix} H_1 \\ \delta A_1 \end{pmatrix} = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix},$$

where

$$R_1 = -\frac{J_1}{L},$$

and

$$R_2 = -\frac{2v_1}{L} \dot{V} + \frac{v_6}{2} \dot{h}_1.$$

The determinant is

$$\Delta_V = 4b_1 v_{10} - v_{11}^2.$$

Hence

$$H_1 = \frac{2v_{10}R_1 - v_{11}R_2}{\Delta_V},$$

and

$$\delta A_1 = \frac{-v_{11}R_1 + 2b_1R_2}{\Delta_V}.$$

The δA_0 equation gives independently

$$\delta A_0 = \frac{(v_1 V)'}{L v_9} - \frac{v_8 h_1 + v_{12} H_2 + v_{13} \delta\phi}{2v_9}.$$

Thus the previously open variables $H_1, \delta A_0, \delta A_1$ are not an unresolved formal step: their exact Schur-complement elimination is known once the common radial coefficient jets are supplied.

In the pure Einstein-plus-genuine-SVT limit, the coefficient identities reduce to

$$b_1 = \alpha_1, \quad v_9 = \alpha_4, \quad v_{10} = \alpha_5, \quad v_{11} = -2\alpha_2.$$

Consequently

$$\Delta_V = 4\alpha_1\alpha_5 - 4\alpha_2^2 = -4C_{\text{odd}},$$

where

$$C_{\text{odd}} = \alpha_2^2 - \alpha_1\alpha_5 > 0.$$

Using the frozen strong-field regression

$$C_{\text{odd},\text{min}} \simeq 0.76072,$$

we obtain the strict pivot bound

$$|\Delta_V| \geq 3.04288.$$

Before dividing the two equations by L , the full determinant carries a factor L^2 . At $\ell=2$, $L=6$, this gives

$$|L^2\Delta_V| \geq 109.54368.$$

The δA_0 pivot is $2Lv_9=2L\alpha_4$. With

$$\alpha_{4,\text{min}} \simeq 9.996 \times 10^{-3},$$

one finds at $\ell=2$

$$2L\alpha_4 \geq 0.119952.$$

Together with

$$\min |D_{h_1}| \simeq 11.99,$$

all nondynamical even-parity eliminations therefore possess finite nonzero pivots in the reference strong-field construction.

20.5 Fully reduced physical even operator

After the four nondynamical variables have been removed in the order

$$h_1 \longrightarrow (H_1, \delta A_1) \longrightarrow \delta A_0,$$

the remaining physical even variables may be collected as

$$\vec{\chi} = (\psi, \delta\phi, V)^T.$$

No additional propagating field is created by the handover: for `\ell1\ge2` the mode count remains the expected three even physical degrees of freedom. The local quadratic action can therefore be written, after integrations by parts, in the standard form

$$S_{\text{even}}^{(2)} = \frac{1}{2} \int dt dr \left[\dot{\vec{\chi}}^T K \dot{\vec{\chi}} + \dot{\vec{\chi}}^T R \vec{\chi}' - \vec{\chi}'^T G \vec{\chi}' - \vec{\chi}^T M \vec{\chi} \right].$$

Here K , R , G , and M are smooth functions of the **unreduced common action jets** and of the frozen P5 background. They are not obtained by adding already-reduced Horndeski and SVT matrices. Algebraically, each elimination is a Schur complement of the corresponding nondynamical block. Schematically, for a nondynamical block N ,

$$\mathcal{Q}_{\text{red}} = \mathcal{Q}_{qq} - \mathcal{Q}_{qN} \mathcal{Q}_{NN}^{-1} \mathcal{Q}_{Nq}.$$

The explicit nonzero pivots established in Secs. 20.3–20.4 guarantee that these Schur complements are smooth on the certified domains.

The radial high-frequency characteristic polynomial is

$$\mathcal{P}_r(\omega, k) = \det(\omega^2 K - \omega k R - k^2 G) = 0,$$

while the angular eikonal symbol is obtained from the leading large- L angular operator after the same constraint reduction. Local ghost freedom requires positive eigenvalues (equivalently positive principal minors in the chosen basis) of K ; radial and angular Laplacian stability require real characteristics with positive squared propagation speeds.

The relevance of the same-action theorem below is now precise. Away from the tested pivots, K , R , G and the angular principal operator are smooth rational functions of a **finite set of action jets**. Strict positivity of their healthy-sector margins is therefore an open condition in jet space. This is what permits a controlled continuation through the two compact overlap bands.

21. Constructive Same-Action Closure

21.1 Meaning of “same-action closure”

The phrase **same-action closure** is used here in a precise local sense. It does not mean that the data determine a unique fundamental Lagrangian on all of field space. It means that there exists one smooth local covariant Horndeski–SVT action, defined on a tubular neighborhood of the one-dimensional frozen P5 background trajectory, whose restrictions reproduce the already constructed exterior Horndeski, strong-field genuine-SVT and inner quartic-Horndeski representatives and whose quadratic principal symbol remains in the locally healthy domain through both handovers.

Write the common action schematically as

$$S_{\text{HSVT}}[g, \phi, A] = \int d^4x \sqrt{-g} \left[\mathcal{L}_H(G_2, G_3, G_4, G_5) + \mathcal{L}_{\text{SVT}}(f_2, f_3, \tilde{f}_3, f_4, \tilde{f}_4) \right],$$

with the shared Einstein/scalar/Maxwell pieces counted once according to the convention of Sec. 5.3. The closure problem is therefore a problem in the finite action-jet bundle sampled by the static equations and the quadratic perturbation action.

At a point of the background curve let

$$\mathcal{J} = (\mathcal{J}_{\parallel}, \mathcal{J}_{\perp})$$

denote the required finite collection of action values and derivatives. $\mathcal{J}_{\parallel}$ contains combinations constrained by the static field equations; $\mathcal{J}_{\perp}$ contains transverse derivatives that vanish from the background equations but enter the perturbation coefficients. The latter are the background-null controls.

The closure established below consists of four ingredients:

1. a regular scalar coordinate $\phi = \xi$ on both overlap bands;
2. a flat $\mathcal{C}^\infty$ partition in that scalar coordinate;
3. nonsingular background/control Jacobians permitting the constrained and stability-sensitive jets to be re-solved smoothly;
4. nonsingular common constraint pivots permitting the physical even operator to be formed throughout the continuation.

These four statements are enough to convert the previously separate healthy local representatives into one local same-action section.

21.2 Scalar-coordinate $\mathcal{C}^\infty$ action partition

On each overlap band choose the endpoint scalar values $\phi_a < \phi_b$ and define

$$t(\phi) = \frac{\phi - \phi_a}{\phi_b - \phi_a}.$$

Because the reconstructed P5 profile is strictly monotone there, ϕ is an admissible local field-space coordinate and no explicit coordinate dependence $\mathbf{r}$ has to be inserted into the fundamental action.

The partition is

$$s_\infty(t) = \frac{e^{-1/t}}{e^{-1/t} + e^{-1/(1-t)}} \quad (0 < t < 1),$$

extended by 0 for $t \leq 0$ and by 1 for $t \geq 1$. It obeys

$$s_\infty^{(n)}(0) = s_\infty^{(n)}(1) = 0 \quad \text{for every } n \geq 1.$$

Consequently, for every finite jet order m used by the perturbation problem, a glued coefficient returns exactly to the pure endpoint jets:

$$\partial_\phi^m \mathcal{J}_{\text{hyb}}(\phi_a) = \partial_\phi^m \mathcal{J}_L(\phi_a),$$

$$\partial_\phi^m \mathcal{J}_{\text{hyb}}(\phi_b) = \partial_\phi^m \mathcal{J}_R(\phi_b).$$

This is the reason the compactly extended quintic Hermite bridge, although variationally optimal for fixed endpoint two-jets, is retained only as a geometric reference. The full action reducer differentiates coefficient combinations beyond second order, so a flat C^∞ handover is the natural same-action choice.

The numerical P5 bands are

$$I_{\text{out}} : \quad 0.5515230871346237 \leq u \leq 0.61,$$

and

$$I_{\text{in}} : \quad 0.71 \leq u \leq 0.715.$$

On I_{out} ,

$$\min f \simeq 0.387706, \quad \min h \simeq 0.426991, \quad \min |\phi'| \simeq 0.446924,$$

and on I_{in} ,

$$\min f \simeq 0.288814, \quad \min h \simeq 0.389107, \quad \min |\phi'| \simeq 1.240374.$$

Thus neither overlap contains a metric zero or a scalar-coordinate turning point.

21.3 Background re-solving rather than blind interpolation

A naive convex interpolation of all action coefficients would not, by itself, guarantee the exact P5 background equations. The construction therefore interpolates only a provisional jet section and then **projects it back onto the background equation manifold**.

Let

$$\mathcal{C}_A(\phi, \mathcal{J}) = 0$$

collect the independent static equations, for example

$$\mathcal{E}_{00} = 0, \quad \mathcal{E}_{11} = 0, \quad \mathcal{E}_\phi = 0, \quad \mathcal{E}_A = 0,$$

after removal of dependent combinations. Choose a set of constrained jet coordinates $Y^A \subset \mathcal{J}_{\parallel}$ whose background Jacobian is nonsingular on the local representative. The implicit-function theorem then gives

$$Y^A = Y^A(\phi, Z),$$

smoothly in the remaining freely specified jets Z , throughout a sufficiently small neighborhood. The compact overlap interval is covered by finitely many such neighborhoods.

Operationally this is the procedure already used in the separate reconstructions: the P5 metric and scalar profile are held fixed, freely selectable background-null data are specified, and the coefficients entering the actual background equations are re-solved. The handover therefore does not perturb the frozen geometry merely to make the action convenient.

21.4 Background-null stability controls and the SVT rank certificate

The decisive freedom is transverse to the one-dimensional background curve. In the genuine-SVT sector select the three background-null Hessian controls

$$(H_{XX}, H_{XF}, H_{FF}).$$

For the stability-sensitive targets (v_1, v_4, c_2) the corresponding control submap contains

$$M_{\text{SVT}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ \kappa/2 & -F_{\text{bg}} & 0 \end{pmatrix}, \quad \kappa = h\phi'^2.$$

Its determinant is

$$\det M_{\text{SVT}} = \frac{\kappa}{2}.$$

This identity is stronger than a numerical rank observation. Since $h > 0$ and $\phi' \neq 0$ on both overlaps, rank three is guaranteed there independently of the background electric invariant.

For $r_s=1$ the numerical rank margins are

domain	$\min \kappa$	$\min \det M_{\text{SVT}} $
I_{out}	0.0988384	0.0494192
$0.61 \leq u \leq 0.715$	0.207862	0.103931
I_{in}	0.632753	0.316377

and no loss of control rank occurs at either handover.

The corresponding full Horndeski principal-control map, formed from nonredundant targets such as

$$(\mathcal{F}, \mathcal{H}, a_1, c_4, c_2),$$

and the available G_i derivatives, retains full rank over the onset-to-core interval in the existing reconstruction. Importantly, the light-ring condition

$$rf' - 2f = 0$$

does not make either control map singular. The photon sphere is therefore a property of the null geometry, not a forced degeneracy of the action-jet reconstruction.

21.5 Exact common constraint invertibility

Background solvability is not enough: the common unreduced even action must also remain reducible. Sections 20.2–20.4 expose all nondynamical pivots relevant to the chosen even basis.

The first constraint gives

$$h_1 = -\frac{a_3\psi' + B_{\text{eff}}\delta\phi' + C_\phi\delta\phi + C_{H_2}\psi + v_2V}{D_{h_1}}.$$

For $\ell=2$,

$$\min |D_{h_1}| \simeq 11.99,$$

and the numerical scan finds no zero through approximately $L=1000$.

After this substitution, the H_1 and δA_1 equations have determinant

$$\Delta_V = 4b_1v_{10} - v_{11}^2.$$

In the pure genuine-SVT limit,

$$b_1 = \alpha_1, \quad v_{10} = \alpha_5, \quad v_{11} = -2\alpha_2,$$

so that

$$\Delta_V = -4(\alpha_2^2 - \alpha_1\alpha_5) = -4C_{\text{odd}}.$$

The odd-sector stability condition $C_{\text{odd}} > 0$ therefore implies the invertibility of this even constraint block. With

$$C_{\text{odd},\min} \simeq 0.76072,$$

one obtains

$$|\Delta_V| \geq 3.04288.$$

Before the equations are divided by L , the determinant contains an additional L^2 . At the lowest gravitational multipole, $\ell=2$ and $L=6$,

$$|L^2 \Delta_V| \geq 109.54368.$$

The ΔA_0 equation has the pivot $2L v_9$. In the same pure-SVT limit $v_9 = \alpha_4$, and the frozen reference margin

$$\alpha_{4,\min} \simeq 9.996 \times 10^{-3}$$

gives

$$2L\alpha_4 \geq 0.119952 \quad (\ell = 2).$$

Thus no nondynamical variable is eliminated by division through a vanishing coefficient in the certified reference construction. In particular, the stable inner light ring does not coincide with a constraint singularity.

21.6 Continuation of the healthy principal symbol

Let $\mathbf{T}(\mathcal{J})$ denote a finite set of stability targets sufficient to characterize the local principal symbol in the chosen representation: odd kinetic/gradient combinations, the even kinetic matrix, radial characteristic coefficients and the leading angular symbol. The already constructed pure representatives have strict margins, including

$$K_{\text{ext},\min} \simeq 9.7423 \times 10^{-3},$$

$$C_{\text{odd},\min} \simeq 0.76072, \quad \alpha_{4,\min} \simeq 9.996 \times 10^{-3}, \quad \alpha_{7,\min} \simeq 2.00153,$$

$$\mathcal{K}_{\text{even},\min} \simeq 10.5415,$$

$$c_{r,T}^2 \geq 1.00507, \quad c_{r,S}^2 \geq 0.903745, \quad c_{r,V}^2 \geq 9.45887,$$

and

$$c_{\Omega,-}^2 \geq 84.912, \quad c_{\Omega,+}^2 \geq 643.142.$$

For the inner quartic-Horndeski core,

$$\mathcal{F}_{\min} = 1, \quad \min(\mathcal{G}, \mathcal{H}) \simeq 0.999381.$$

These are strict inequalities, not zeros. Away from the pivots of Sec. 21.5 the reduced principal matrices are smooth functions of $\mathcal{J}$. Therefore the healthy set

$$\mathcal{U}_+ = \{\mathcal{J} : K > 0, c_r^2 > 0, c_\Omega^2 > 0, \text{ constraints invertible}\}$$

is open in the finite jet topology.

The control maps of Sec. 21.4 allow the background-null jets to be adjusted while the background-constrained jets evolve through a handover. Choose any smooth path $\mathbf{T}_*(\phi)$ in the positive target domain connecting the endpoint target values. For positive scalar targets this can be taken inside the positive orthant; for positive-definite kinetic blocks the positive-definite cone is convex, so a smooth positive path exists. The full-rank control Jacobian then permits the transverse jets to be solved locally so that

$$\mathbf{T}(\mathcal{J}(\phi)) = \mathbf{T}_*(\phi).$$

Because each handover interval is compact, the local implicit-function neighborhoods possess a finite subcover. Continuing from one neighborhood to the next constructs a smooth healthy jet section across the full band.

This argument is the step that turns “healthy endpoints” into a **same-action local existence result**. It does not assume that arbitrary interpolation preserves stability; it actively re-solves the available transverse controls to remain inside $\mathscr{U}_+$.

21.7 Realization as one local covariant action

A smooth jet section along the background curve still has to be realized by ordinary functions, rather than being left as a table of pointwise derivatives. This is provided constructively by the tubular expansion introduced in Sec. 16.3.

Let n^A be smooth normal coordinates transverse to the P5 trajectory in invariant space. For each independent action function write, to the finite order needed by the quadratic problem,

$$F(\phi, n) = F_0(\phi) + A_A(\phi)n^A + \frac{1}{2}H_{AB}(\phi)n^An^B + \frac{1}{3!}T_{ABC}(\phi)n^An^Bn^C + \dots$$

The coefficient functions are precisely the smooth same-action jet section constructed above. Multiplying the normal expansion by a smooth tubular cutoff that equals unity on the background curve yields a genuine local smooth function of the covariant invariants. Repeating this for the finite set of $\mathbf{G}_i$, $\mathbf{f}_i$ and $\tilde{\mathbf{f}}_i$ functions produces one local covariant action, not a radius-by-radius prescription.

Because $\mathbf{s}_\infty$ is flat at each patch boundary, the resulting action agrees with the selected pure representatives to every finite derivative order used in the calculation. The exterior, strong-field and core descriptions are therefore different coordinate patches/representatives of the same local action construction rather than dynamically disconnected theories.

No uniqueness is implied. Background reconstruction is underdetermined, and the background-null directions generate a family of admissible same-action completions.

21.8 Local Same-Action Closure Theorem

The preceding construction may be summarized as a theorem tailored to the frozen P5 background.

Theorem (constructive local same-action closure). Let the frozen P5 background satisfy the source-locked geometry of Sec. 3, and let the exterior Horndeski, strong-field genuine-SVT and inner quartic-Horndeski representatives be the locally healthy sections constructed in Secs. 6–14. Assume on the two finite overlaps that:

$$f > 0, \quad h > 0, \quad \phi' = \Xi' \neq 0;$$

that the background equation Jacobian for the chosen constrained jets is nonsingular; that the Horndeski and SVT transverse control maps have full rank; and that the common nondynamical pivots $D_{\{h_1\}}$, $2L_v_9$ and Δ_V do not vanish. Then there exists a C^∞ local covariant Horndeski–SVT action on a tubular neighborhood of the complete P5 background trajectory such that:

$$\mathcal{E}_{\mu\nu} = 0, \quad \mathcal{E}_\phi = 0, \quad \mathcal{E}_A = 0$$

on the frozen background; the chosen pure representatives and all finite endpoint jets are recovered outside the transition bands; the common even constraints are smoothly eliminable; and the local odd/even principal symbol can be continued through both transitions without crossing a ghost, Laplacian, hyperbolicity or constraint-singularity boundary.

Proof. Monotonicity of ϕ supplies a scalar-coordinate chart on each overlap. The flat partition s_∞ produces smooth provisional action jets and exact finite-jet endpoint recovery. The non-singular background Jacobian re-solves the constrained jets so that the frozen P5 fields continue to satisfy the static equations. The full-rank transverse control maps re-solve the background-null jets so that the chosen stability targets follow a smooth path inside the strict healthy domain. The nonzero constraint pivots make every Schur complement in the common even reduction smooth. Strict positivity of the kinetic and characteristic margins is an open condition, and compactness of each overlap gives a finite continuation cover. Finally, the tubular normal expansion realizes the resulting smooth finite jet section as ordinary local action functions. QED.

The theorem is local in field space and along the static P5 branch. It is nevertheless the precise closure needed to remove the previous “patchwork action” objection at the level addressed by the present paper.

21.9 Machine-readable closure certificate

The construction was encoded in an independent machine-readable certificate with eight explicit gates:

G_1	$f > 0, h > 0$ on both handover bands
G_2	$\phi = \Xi$ monotone with $\phi' \neq 0$
G_3	$\det M_{\text{SVT}} = \kappa/2 \neq 0$
G_4	s_∞ monotone and C^∞ -flat
G_5	all frozen pure-sector local stability margins strict
G_6	$2Lv_9 \neq 0$
G_7	$\Delta_V \neq 0$
G_8	$D_{h_1} \neq 0$

All eight gates pass in the 14 September 2026 reference run. The machine status is

PASS_CONSTRUCTIVE_LOCAL_SAME_ACTION_CLOSURE_CERTIFICATE.

The numerical band/rank data are:

band	$\Delta\Xi$	$\min f$	$\min h$	$\min \det M_{\text{SVT}} $
I_{out}	0.0960785	0.387706	0.426991	0.0494192
I_{in}	0.0124100	0.288814	0.389107	0.316377

and the full genuine-SVT lobe has

$$\min h \simeq 0.326857, \quad \min |\det M_{\text{SVT}}| \simeq 0.103931.$$

The certificate is an existence/regularity certificate. It should not be confused with a unique parameter fit.

21.10 What is now closed, and what remains outside the closure claim

The following items are **closed within the scope of this paper**:

- the P5 geometry is source-locked and independently reconstructible;
- the exterior, light-ring and core action representatives overlap;
- the transition is expressible in scalar field space with a flat $\mathcal{C}^\infty$ partition;
- the background equations can be re-solved rather than violated by the handover;
- the SVT transverse stability-control map has analytic rank three;
- the full Horndeski principal-control map is nondegenerate on the certified interval;
- $h_1, H_1, \delta A_0$, and δA_1 are algebraically eliminable with nonzero reference pivots;
- the local odd/even principal-symbol health conditions can be continued through both overlaps;
- the resulting smooth finite jet section can be realized by one local covariant action.

The following statements remain **outside** the closure theorem:

- uniqueness or microscopic fundamentality of that action;
- a single preferred closed-form choice for every free function over all field space;
- an archived full K, R, G, M matrix dump for one arbitrarily selected member of the certified family;
- the complete finite- ℓ quasinormal spectrum and the lifetime of stable-trapped modes;

- nonlinear stability under finite perturbations;
- proof that every effective characteristic cone lies inside the matter light cone;
- a UV completion or a theorem excluding every nonperturbative pathology.

This distinction is essential. The **local same-action closure is complete**; the global dynamical and microscopic interpretation of the resulting theory is not.

The internal SSZ Physics RAG was used as a provenance cross-check for the canonical Ξ hierarchy, the historical variational bridge theorem, and the August dynamics-gap audit. The new September P5 action-jet and closure-certificate numbers are treated as current project regressions rather than retroactively labelled as RAG-independent validations.

22. Interpretation of the Large Characteristic Speeds

Several vector and angular characteristic speeds in the constructed SVT representative substantially exceed unity.

It is important not to conflate three different statements.

A negative squared speed,

$$c^2 < 0,$$

signals a Laplacian instability.

A zero speed can signal a degeneracy or strong-coupling issue depending on the surrounding kinetic structure.

A positive super-unit speed,

$$c^2 > 1,$$

does neither automatically.

The present result establishes only the first level of consistency:

$$c^2 > 0.$$

A complete physical interpretation requires identifying the relevant causal cone of the interacting gravitational–scalar–vector system and determining whether the effective theory admits a consistent UV interpretation.

In modified-gravity systems, different modes need not propagate on the matter metric $g_{\{\mu\nu\}}$.

Their principal operators define effective metrics

$$\mathcal{G}_{(i)}^{\mu\nu}$$

whose null cones can differ.

Whether superluminal characteristics imply a fundamental inconsistency depends on additional properties including global causal structure, analyticity, UV completion assumptions, and the operational definition of signal propagation.

We therefore make no claim of causal completion in this paper.

The appropriate terminology for the established result is:

local ghost/Laplacian/hyperbolicity stability.

23. Stable Trapping and the Difference Between Local and Global Stability

The inner stable light ring deserves separate emphasis.

A healthy local principal symbol does not guarantee rapid decay of perturbations.

Stable trapping can prevent efficient dispersal.

A wave packet concentrated near the stable null orbit may repeatedly return to the trapping region.

Even in a locally hyperbolic theory, this can produce slowly decaying resonances or unusually long-lived quasinormal modes.

The shallow depth of the P5 null well,

$$\Delta_{\text{well}} \simeq 0.42\%,$$

may soften this effect, but it does not remove the underlying mathematical question.

The correct next observable calculation after local closure is therefore a finite-\ell spectral analysis.

One should compute the coupled tensor-scalar-vector quasinormal spectrum of the final hybrid action and determine whether the stable orbit produces:

$$\text{Im } \omega \rightarrow 0^-,$$

an actual growing mode

$$\text{Im } \omega > 0,$$

or merely a finite family of long-lived but decaying resonances.

This question cannot be answered from the eikonal local positivity tests alone.

24. Nonlinear Stability as a Separate Problem

Even a complete linear mode analysis would not prove nonlinear stability.

The full hybrid equations can schematically be written as

$$\mathcal{E}_A [\Psi, \partial\Psi, \partial^2\Psi] = 0,$$

where Ψ collectively denotes

$$g_{\mu\nu}, \quad \phi, \quad A_\mu.$$

A nonlinear stability theorem would require an appropriate gauge choice, a hyperbolic formulation of the evolution system, control of the constraint propagation, and energy estimates capable of bounding suitable Sobolev norms of the perturbations.

The relevant logical implication would have the form

$$E_N(t) < \infty \implies \text{the solution can be continued.}$$

One would then seek estimates such as

$$\frac{dE_N}{dt} \leq C(E_N)E_N.$$

A sufficiently strong bound could rule out finite-time blow-up for sufficiently small perturbations of the P5 background.

Nothing in the present local principal-symbol analysis is presented as a substitute for such a theorem.

Rather, the present work establishes the prerequisite that the linearized local equations possess a healthy characteristic structure in the first place.

25. Discussion: P5 in the Landscape of Compact-Object Geometries

25.1 Comparison with Schwarzschild general relativity

The P5 closure is deliberately close to Schwarzschild in the regimes where present observations are most constraining, but it is not globally Schwarzschild.

In the weak field,

$$f = 1 - u + O(u^4), \quad h = 1 - u + O(u^4),$$

so the first displayed P5 deviations occur only at fourth order in the present compactness expansion.

At the outer light ring, the agreement is considerably stronger than equality of the coordinate radius alone. The P5 photon condition reduces to

$$Q(u) \equiv \frac{u\chi}{1 + \Xi} - 1 = 0.$$

For the outer root,

$$u = \frac{2}{3}, \quad x = \frac{3}{2},$$

we obtain numerically

$$\Xi = \sqrt{3} - 1,$$

$$\chi = \frac{3\sqrt{3}}{2} = \chi_{\text{GR}},$$

and consequently

$$\boxed{f = h = \frac{1}{3}}.$$

Define the dimensionless optical potential

$$W(u) = u^2 f(u).$$

Since

$$\frac{d \ln W}{du} = -\frac{2Q}{u},$$

one has at a light ring

$$W'' = -\frac{2W}{u}Q'.$$

The source-locked P5 closure gives at the outer orbit

$$Q' \simeq 4.5, \quad \boxed{W'' \simeq -2},$$

which is the Schwarzschild value in the same dimensionless coordinate. Thus the outer optical calibration simultaneously matches the radius, $\mathbf{f}$, $\mathbf{h}$, susceptibility, and local curvature of the null potential to numerical precision. This explains why the shadow scale and eikonal instability scale can be Schwarzschild-like even though the interior geometry is qualitatively different.

The departure from GR is concentrated inward. Schwarzschild reaches $\mathbf{f}=\mathbf{h}=0$ at $\mathbf{u}=1$; P5 instead tends to

$$f \rightarrow \frac{1}{4}, \quad h \rightarrow 1,$$

and contains a second, stable light ring before entering the regular core.

25.2 P5 is not a regular black hole in the strict causal sense

The phrase “regular black hole” is often used for nonsingular metrics that preserve an event horizon while replacing the Schwarzschild singularity by a regular core, frequently a de Sitter-like region. Many such models possess both an outer event horizon and an inner Cauchy horizon; the latter raises the separate mass-inflation stability problem.

The frozen P5 closure belongs to a different causal class. The global scan gives

$$f \geq \frac{1}{4},$$

and

$$h \geq 0.3268566652$$

through the source-locked domain and its regular-core continuation. There is no zero of either metric function and hence no event horizon or inner Cauchy horizon of the usual static spherical type.

P5 therefore avoids the standard inner-horizon mass-inflation problem by not possessing that horizon structure in the first place. This should not be advertised as a generic stability advantage over regular black holes, because P5 acquires a different potential problem: stable trapping at the inner light ring.

25.3 Relation to horizonless ultracompact objects

A horizonless object with a light ring belongs naturally to the ultracompact-object class. In broad metric theories, light-ring structure is strongly constrained topologically; under additional Einstein-equation and energy-condition assumptions, smooth horizonless ultracompact objects are known to develop light rings in pairs with a stable member.

P5 exhibits precisely the geometric pair

$$x_{\text{out}} = 1.5$$

and

$$x_{\text{in}} \simeq 1.41616064.$$

One should nevertheless distinguish the general geometric pairing argument from the GR-specific energy-condition theorem. The final P5 support is a Horndeski–SVT effective stress system, so conclusions relying specifically on the Einstein equations with ordinary matter cannot simply be imported unchanged.

The useful comparison is therefore structural rather than identificational: P5 occupies the horizonless ultracompact sector, but with a covariantly reconstructed scalar–vector–tensor support rather than a prescribed GR material star.

25.4 Stable trapping: the principal unresolved physical risk

Stable trapping is the strongest reason not to equate local hyperbolicity with global viability.

For wave equations on spacetimes with stable trapping, rigorous results show that uniform decay can be no faster than logarithmic in relevant classes. In ultracompact-object models the stable null orbit is also associated with very long-lived modes, and nonlinear numerical evolutions of horizonless bosonic stars have exhibited light-ring-triggered migration or collapse.

The P5 case has an important quantitative peculiarity: its null well is shallow,

$$\Delta_{\text{well}} \simeq 4.18 \times 10^{-3}.$$

Nonlinear ultracompact-object studies indicate that very shallow stable-light-ring wells can substantially lengthen the instability timescale. That is encouraging but not exculpatory. It means the physically decisive calculation is now a **timescale calculation**, not another local sign test.

After same-action closure, the next required program is therefore:

$$\text{finite-}\ell \text{ spectrum} \rightarrow \text{quasinormal/resonant lifetimes} \rightarrow \text{nonlinear evolution.}$$

A viable outcome would be either the absence of growing modes with sufficiently rapid decay, or trapping timescales so long that the configuration remains astrophysically relevant. A rapidly growing finite- ℓ or nonlinear mode would instead falsify the proposed stationary P5 compact object even though all local ghost and Laplacian conditions are positive.

25.5 Relation to the SSZ interpretation

The covariant reconstruction should also be distinguished from the conceptual motivation of Segmented Spacetime.

SSZ begins from a geometrical/operational interpretation in which phenomena conventionally encoded as spacetime curvature are organized through a primary segmentation variable Ξ , with

$$D = (1 + \Xi)^{-1}.$$

The present P5 calculation strengthens that connection: the temporal metric remains exactly Ξ -primary, while the radial geometry is generated by the susceptibility closure $\chi(\Xi)$ rather than by an independent fitted metric function.

At the same time, the paper deliberately does **not** require the reader to accept a microscopic interpretation of segmentation. The covariant question is narrower:

Can the source-locked P5 geometry be embedded into a local field theory whose background equations and perturbative principal symbol are healthy?

This separation is useful because it leaves three logically independent layers:

1. the **SSZ interpretation** of Ξ ;
2. the **P5 geometrical closure** $\chi(\Xi)$;
3. the **Horndeski–SVT dynamical realization** of that geometry.

Agreement or failure at one layer need not be disguised as agreement or failure at another. In particular, the old SSZ Hermite bridge remains a mathematically justified interpolation template, while the present Horndeski–SVT action is the object that must ultimately determine the physical dynamics.

26. Main Results

The analysis produces a closed local dynamical embedding of the frozen P5 geometry.

The geometry remains fixed throughout. Its outer unstable light ring is

$$x = 1.5,$$

and its stable inner ring is

$$x \simeq 1.41616064.$$

The outer orbit reproduces the Schwarzschild temporal/radial metric values and null-potential curvature at the calibrated point, while the inner ring generates a shallow stable-trapping well. The global P5 continuation remains horizonless with positive $\mathfrak{f}$ and $\mathfrak{h}$.

The exterior admits a finite Horndeski scalar carrier with positive kinetic margin. The complete two-light-ring domain admits a background-integrable genuine $\mathcal{U}(1)$ SVT action section. Its scalar equation is solved explicitly through the reconstruction of $\mathfrak{f}_{\{3,\mathbf{X}\}}$. The odd sector has positive kinetic and gradient margins; the corrected even kinetic quantity obeys

$$\mathcal{K} \geq 10.5415,$$

and the independent $\mathfrak{e}_1$ identity is satisfied at the 10^{-10} level. The scalar radial characteristic remains near unity,

$$0.903745 \lesssim c_{r,S}^2 \lesssim 1.0614,$$

while the tensor, vector and coupled angular squared characteristics remain positive. The regular quartic-Horndeski core overlaps the genuine-SVT lobe.

The action-level handover is constructed in the monotone scalar coordinate $\phi=\mathbf{X}$ with a flat $\mathcal{C}^\infty$ partition. Background-constrained jets are re-solved from the static equations; background-null transverse jets are used to control the perturbative principal symbol. The selected SVT control map has the analytic determinant

$$\det M_{\text{SVT}} = \frac{1}{2} h \phi'^2,$$

which is strictly nonzero on both overlap bands.

The common even-parity reduction is also complete at the algebraic level. The generalized variable

$$\psi = H_2 + \frac{La_4}{a_3}h_1 + \frac{a_1}{a_3}\delta\phi'$$

unifies the Horndeski and genuine-SVT constraint structures. The H_0 constraint yields h_1 with

$$\min |D_{h_1}| \simeq 11.99$$

for the reference $\ell=2$ scan. The remaining H_1 - δA_1 block has

$$\Delta_V = 4b_1v_{10} - v_{11}^2,$$

and in the pure-SVT limit

$$\Delta_V = -4C_{\text{odd}},$$

so that

$$|\Delta_V| \geq 3.04288.$$

The δA_0 pivot obeys

$$2L\alpha_4 \geq 0.119952$$

at $\ell=2$. None of the exposed nondynamical constraints becomes singular at either light ring or either handover.

Combining the nonzero background/control ranks, the strict pure-sector margins, the smooth constraint reduction and the flat scalar-coordinate partition gives the constructive Local Same-Action Closure Theorem of Sec. 21. Its machine-readable eight-gate certificate returns

PASS_CONSTRUCTIVE_LOCAL_SAME_ACTION_CLOSURE_CERTIFICATE.

The principal result is therefore stronger than the statement that several healthy patches exist: **one smooth local covariant Horndeski–SVT action can be constructed through the full P5 background trajectory, including both light rings and both action handovers, without a forced local ghost, Laplacian, hyperbolicity or constraint singularity.**

27. What Is Not Proven by the Closure

The local closure theorem deliberately stops short of several stronger claims.

It does not select a unique fundamental action. The inverse reconstruction is underdetermined, and the background-null directions generate a family of same-action completions.

It does not provide one privileged elementary closed-form expression for every free G_i , f_i , and $\tilde{f}_i$ over all of invariant field space. The theorem constructs a smooth local action in a

tubular neighborhood of the P5 trajectory, which is the domain needed for the background and its local perturbations.

It does not replace an implementation-level archive of the complete reduced K, R, G, M matrices for a chosen representative. Producing that matrix dump and rerunning endpoint/pure-limit regressions remains valuable for independent reproducibility, but it is no longer an existence obstruction.

It does not establish that every positive effective characteristic speed is subluminal relative to the matter cone. In fact several vector/angular characteristics are substantially super-unit in the chosen normalization, so causal-cone and EFT interpretation remain separate questions.

It does not compute the complete finite- ℓ quasinormal spectrum. Therefore it does not determine the lifetime or nonlinear significance of the shallow stable-trapping well.

It does not prove nonlinear stability, global well-posedness for arbitrary finite data, the absence of every nonperturbative pathology, or a UV completion.

The post-closure distinction is therefore

local same-action existence and principal-symbol regularity: **CLOSED**

versus

global spectral, nonlinear, causal and microscopic completion: **OPEN**.

28. Post-Closure Program

The next calculation is no longer another attempt to make the Horndeski and SVT patches meet. They already meet at the level of the local action and principal symbol.

The first post-closure task is **representative-level reproducibility**: select one explicit member of the certified same-action family, generate the complete radial arrays for the unreduced coefficients and the reduced K, R, G, M matrices, and archive the full pure-Horndeski, pure-SVT and transition regressions. This will turn the existence certificate into a convenient benchmark implementation.

The second task is the finite-multipole spectral problem. The coupled perturbation equations should be solved with physically appropriate boundary conditions across the full horizonless geometry, yielding the quasinormal and trapped-mode spectrum. The stable inner light ring is the central target, because a positive local principal symbol does not determine the lifetime of modes trapped in a shallow potential well.

The third task is nonlinear evolution. Time-domain simulations should test whether the stable-trapping region drives migration, collapse, long-lived oscillations, or remains benign on astrophysically relevant timescales.

The fourth task is the causal/EFT interpretation of the characteristic cones. The effective scalar, vector and tensor cones should be compared with the matter light cone, and the large positive vector/angular speeds should be analyzed in terms of hyperbolicity, front velocity, cutoff scale and possible UV completion rather than being conflated with a Laplacian instability.

Finally, rotation should be incorporated. A slowly rotating extension would test whether the two-light-ring structure and the same-action control rank persist once spherical symmetry is relaxed,

before attempting a fully stationary axisymmetric construction.

29. Conclusion

We investigated the dynamical realizability of the frozen P5 Segmented-Spacetime geometry within a covariant Horndeski–scalar–vector–tensor framework while refusing to alter the geometry merely to fit a convenient theory.

The P5 background is Ξ -primary in its temporal sector and is closed radially by the source-locked susceptibility relation. It is horizonless and regular, contains an outer unstable light ring at

$$x = 1.5,$$

and an inner stable light ring at

$$x \simeq 1.41616064.$$

A finite exterior Horndeski carrier, a background-integrable genuine-SVT light-ring carrier and a regular quartic-Horndeski core were constructed with overlapping radial domains. The genuine-SVT lobe passes the employed local odd/even ghost and Laplacian tests with strict margins.

The central closure problem was then treated at the **unreduced action level**. The common generalized variable

$$\psi = H_2 + \frac{La_4}{a_3}h_1 + \frac{a_1}{a_3}\delta\phi'$$

provides the shared constraint coordinate. The h_1 constraint is regular, and the remaining H_1 , δA_0 , and δA_1 variables possess exact nonsingular elimination formulas. In particular,

$$\Delta_V = -4C_{\text{odd}}$$

in the pure genuine-SVT limit, so the already positive odd-sector determinant supplies an analytic lower bound on the even constraint pivot.

The overlap itself is implemented in scalar field space with a flat $\mathbb{C}^\infty$ partition. The frozen background equations are re-solved during the continuation rather than violated by naive interpolation. Background-null transverse action derivatives supply the required perturbative controls; the selected SVT control submap has

$$\det M_{\text{SVT}} = \frac{1}{2}h\phi'^2 \neq 0$$

on both handovers. Since the common reduced operator is smooth away from the tested pivots and all endpoint stability margins are strict, the transverse controls can be continued through the compact overlap bands while remaining inside the healthy principal-symbol domain. A tubular invariant-space expansion then realizes those smooth jets as one ordinary local covariant action.

This establishes the result that was previously missing:

For the frozen P5 background there exists a smooth local covariant same-action Horndeski–SVT realization across the exterior, both strong-field light rings, and the regular core. The static background equations can be satisfied throughout both handovers, every nondynamical even-parity pivot exposed by the common reduction remains nonzero on the certified reference construction, and the local odd/even principal symbol can be continued without a forced ghost, Laplacian, hyperbolicity or constraint singularity. The local same-action closure is therefore complete.

This conclusion is deliberately local and constructive. It does not identify a unique microscopic theory, and it does not turn local hyperbolicity into a global stability theorem. The scientifically important open questions now begin **after** closure: finite-\ell quasinormal modes and stable trapping, nonlinear dynamics, effective causal cones, rotation and UV completion.

Appendix A. Numerical Reference Values for the Frozen P5 Construction

A.1 Explicit P5 closure reference values

In addition to the perturbative reference values below, the frozen geometry itself is now reproducible from

$$\chi(\Xi) = (1 - \Xi)^{3/2} \mathcal{P}_5(\Xi),$$

$$f = (1 + \Xi)^{-2},$$

and

$$h = 1 - \left(\frac{\chi}{\chi_{\text{GR}}} \right)^{2/3} (1 - f), \quad \chi_{\text{GR}} = \frac{1}{2}(1 + \Xi)^3.$$

The key geometry regression points are:

Point	u	x	\Xi	\chi	f	h
natural SVT onset	0.5515230871	1.813161	0.509933	1.469286	0.438616	0.494833
outer light ring	0.6666666667	1.500000	0.732051	2.598076	0.333333	0.333333
stable light ring	0.7061345809	1.4161606400	0.838419	2.603497	0.295877	0.374130
core handover	0.7100000000	1.408451	0.848354	2.534377	0.292705	0.389107

The radial function has a global minimum in the reconstructed closure of approximately

$$h_{\min} = 0.3268566652$$

near $x \simeq 1.473439$, while $f \rightarrow 1/4$ and $h \rightarrow 1$ at the regular center.

The following values define the current frozen numerical reference point of the manuscript.

The natural SVT onset is

$$u_{\text{on}} = 0.5515230871346237,$$

corresponding to

$$x_{\text{on}} \simeq 1.813161.$$

The outer unstable light ring is

$$x_{\text{out}} = 1.5, \quad u_{\text{out}} = \frac{2}{3}.$$

The inner stable light ring is

$$x_{\text{in}} \simeq 1.41616064, \quad u_{\text{in}} \simeq 0.706135.$$

The genuine-SVT integration domain is

$$0.61 \leq u \leq 0.715,$$

or

$$1.3986014 \lesssim x \lesssim 1.6393443.$$

The quartic-Horndeski core reaches outward to

$$x_{\text{core}} = 1.4084507042, \quad u_{\text{core}} = 0.71.$$

The SVT-core overlap is therefore approximately

$$1.39860 \lesssim x \lesssim 1.40845.$$

The scalar background residual of the integrated SVT representative is

$$\max |\mathcal{E}_\phi| \sim 10^{-12}.$$

The reconstructed cubic SVT derivative satisfies

$$\max |f_{3,X}| \simeq 44.73$$

over the extended lobe.

The corrected even no-ghost quantity satisfies

$$10.5415 \lesssim \mathcal{K} \lesssim 137.879.$$

Its independent identity residual is

$$\max \left| \mathcal{K} - e_1 \sqrt{f h} \phi'^2 \right| \simeq 2.18 \times 10^{-10}.$$

The radial characteristic minima are

$$c_{r,T,\min}^2 \simeq 1.00507,$$

$$c_{r,S,\min}^2 \simeq 0.903745,$$

and

$$c_{r,V,\min}^2 \simeq 9.45887.$$

The scalar radial maximum is

$$c_{r,S,\max}^2 \simeq 1.06133.$$

The coupled angular minima are

$$c_{\Omega,-,\min}^2 \simeq 84.912,$$

and

$$c_{\Omega,+, \min}^2 \simeq 643.142.$$

The first coupled angular stability margin is approximately

$$738.609,$$

the second approximately

$$6.0303 \times 10^4,$$

and the minimum discriminant approximately

$$2.9938 \times 10^5.$$

The odd kinetic combination satisfies

$$C_{\text{odd,min}} \simeq 0.76072.$$

The minimum odd coefficients include

$$\alpha_{4,\text{min}} \simeq 9.996 \times 10^{-3},$$

and

$$\alpha_{7,\text{min}} \simeq 2.00153.$$

The regular core satisfies approximately

$$\mathcal{F}_{\text{min}} = 1,$$

$$\mathcal{G}_{\text{min}} = \mathcal{H}_{\text{min}} \simeq 0.999381.$$

The proper central Einstein-frame scalar kinetic coefficient is approximately

$$95.89.$$

The corrected restricted exterior Horndeski scalar kinetic carrier has

$$K_{\text{ext}} \simeq 9.7423 \times 10^{-3}.$$

The first common jet-aware constraint pivot has, for $\ell=2$,

$$\min |D_{h_1}| \simeq 11.99,$$

and no zero or sign change has been observed throughout the tested high-L range.

These numerical values should be regarded as the frozen reference values against which later same-action hybrid calculations are to be regressed.

Appendix B. Reproducibility and Regression Logic

A central methodological principle of the present project is that every new layer of the calculation should reproduce an independently known limit before being used to make a new physical claim.

The current regression chain is

background profile $\rightarrow$ background field equations $\rightarrow$ action jets $\rightarrow$ unreduced quadratic coefficients $\rightarrow$ reduced k

The most important completed internal checks include

$$a_3 + ra_4 = 0$$

for the pure-SVT representative,

$$b_4 - \frac{2ra_4}{f} = 0,$$

$$b_5 + \frac{a_4}{f} = 0,$$

$$d_1 - \frac{a_4}{2f} = 0,$$

and

$$v_7 - \frac{A_0'^2 v_1}{4} = 0$$

to numerical precision.

Independent reconstructions of v_1 , v_4 , and c_2 agree with their stored target values at approximately the 10^{-8} – 10^{-7} level or better on the numerical interpolation grid.

Most importantly,

$$\mathcal{K} = e_1 \sqrt{f h \phi'^2}$$

now closes after correcting the exact P_1 definition.

The future common hybrid reducer will not be accepted as physically validated merely because its matrix algebra is internally consistent.

It must reproduce, in its pure-SVT limit,

$$c_{r,T}^2, \quad c_{r,S}^2, \quad c_{r,V}^2,$$

and the corresponding angular branches obtained from the established SVT formalism.

It must independently reproduce the Maxwell–Horndeski characteristic conditions when the genuine-SVT interactions are switched off.

Only after both regression limits pass will interpolation results be interpreted as physical predictions of the hybrid action.

B.1 Geometry-level regression gates

The following geometry tests are now included before any action reconstruction is accepted:

$$\max |f - (1 + \Xi)^{-2}| \lesssim 6.7 \times 10^{-16},$$

$$\max |h - h[\Xi, \chi]| \lesssim 8.9 \times 10^{-16},$$

and

$$\max |\chi_{\text{stored}} - (1 - \Xi)^{3/2} \mathcal{P}_5(\Xi)| \lesssim 3.6 \times 10^{-15}.$$

The photon equation

$$Q(u) = \frac{u\chi}{1 + \Xi} - 1$$

has exactly two resolved roots in the tested global P5 profile:

$$u_1 = 0.666666666668, \quad u_2 = 0.70613458091.$$

At the outer root the independent optical-potential regression gives

$$W'' \simeq -2$$

with sub- 10^{-9} numerical error in the regenerated profile.

B.2 Variational handover regression

For the normalized flat-two-jet partition,

$$s_{H5} = 10t^3 - 15t^4 + 6t^5,$$

we obtain exactly

$$J_3 = \int_0^1 (s_{H5}''')^2 dt = 720.$$

The independently evaluated $\mathcal{C}^\infty$ bump partition gives

$$J_3 \simeq 2454.648507,$$

so

$$J_3[C^\infty]/J_3[H5] \simeq 3.409234.$$

This regression reproduces the variational ordering expected from the historical SSZ bridge theorem while keeping the bridge outside the fundamental dynamics.

Appendix C. Literature Framework and Provenance Boundaries

The full $\mathcal{L}_2$ – $\mathcal{L}_5$ Horndeski definitions used in Sec. 5 follow the generalized-Galileon representation of Kobayashi, Yamaguchi and Yokoyama, which is equivalent to Horndeski theory up to the standard mappings and integrations by parts.

The explicit second-order $U(1)$ -gauge-invariant scalar–vector–tensor interactions $f_2, f_3, \tilde{f}_3, f_4$, and $\tilde{f}_4$ follow the gauge-invariant SVT construction used by Heisenberg and Tsujikawa. The odd-parity general-SVT stability framework is associated with Heisenberg, Kase and Tsujikawa, while the even-parity general-SVT conditions used for the P5 lobe follow Zhang and Kase.

The Maxwell–Horndeski pure-limit regression follows Kase and Tsujikawa. Their $\ell \geq 2$ decomposition contains two physical odd modes and three physical even modes, matching the mode count used in the common hybrid reducer.

The stable-trapping discussion is intentionally separated from the local ghost/Laplacian result. Keir establishes logarithmic limitations on uniform wave decay in relevant stably trapped spherical geometries. Cardoso *et al.* connect stable null orbits in ultracompact objects with long-lived modes and possible nonlinear problems. Cunha, Berti and Herdeiro establish strong light-ring pairing results for smooth ultracompact objects, with the GR+NEC assumptions carefully distinguished from the more general topological part. Fully nonlinear bosonic-star evolutions by Cunha *et al.* show light-ring-triggered migration or collapse and also emphasize that very shallow wells can lengthen the relevant timescale.

The comparison with regular black holes follows the review by Bonanno and Saueressig: many standard regular-black-hole geometries retain an outer event horizon and an inner Cauchy horizon. Since the frozen P5 closure has $f > 0$ and $h > 0$ everywhere in the tested global continuation, it belongs instead to the horizonless ultracompact class.

The SSZ-specific provenance is deliberately kept separate from established external theory. The canonical SSZ baseline ξ_{weak} , ξ_{strong} , D and the historical C^2 transition are documented in the SSZ complete-documentation corpus, with `SSZ_CANONICAL_FORMULAS_2026.md` used as the later source hierarchy where older branch-lock files conflict. The fixed-two-jet variational theorem is documented in `SSZ_Variational_Bridge_Paper_Wrede_Casu_BingsiAI_v2.pdf`; the warning that a frozen bridge does not uniquely determine dynamics is documented in `SSZ_Middle_Bridge_Dynamics_Gap_Audit_2026-08-29.md`. Neither internal source is treated as external validation of P5. They provide model provenance, a tested interpolation theorem, and a clear separation between geometry and dynamics.

Appendix D. Same-Action Closure Certificate

D.1 Certified overlap data

The machine-readable reference run uses $\mathbf{r_s}=1$ for the numerical $\backslash\text{phi}$ ' and $\backslash\text{kappa}$ normalization. The two same-action transition bands are

$$I_{\text{out}} : \quad 0.5515230871346237 \leq u \leq 0.61,$$

and

$$I_{\text{in}} : \quad 0.71 \leq u \leq 0.715.$$

The corresponding scalar and rank data are

	I_{out}	I_{in}
Ξ_a	0.5099330950	0.8483535179
Ξ_b	0.6060115821	0.8607634713
$\min f$	0.3877061179	0.2888136109
$\min h$	0.4269907185	0.3891074915
$\min \phi' $	0.4469239959	1.2403742162
$\min \kappa$	0.0988383730	0.6327532291
$\min \det \mathbf{M}_{\text{SVT}} $	0.0494191865	0.3163766145

For the full genuine-SVT lobe $0.61 \leq u \leq 0.715$,

$$\min f \simeq 0.28881361, \quad \min h \simeq 0.32685689,$$

$$\min \kappa \simeq 0.20786197, \quad \min |\det \mathbf{M}_{\text{SVT}}| \simeq 0.10393098.$$

D.2 Frozen stability margins used by the certificate

quantity	frozen lower margin
K_{ext}	9.7423×10^{-3}
C_{odd}	0.76072
α_4	9.996×10^{-3}
α_7	2.00153
$\mathcal{K}_{\text{even}}$	10.5415
$c_{r,T}^2$	1.00507
$c_{r,S}^2$	0.903745
$c_{r,V}^2$	9.45887
$c_{\Omega,-}^2$	84.912
$c_{\Omega,+}^2$	643.142
$\mathcal{F}_{\text{core}}$	1
$\min(\mathcal{G}_{\text{core}}, \mathcal{H}_{\text{core}})$	0.999381

The common/pure constraint bounds entering the same certificate are

$$\min |D_{h_1}|_{\ell=2} \simeq 11.99,$$

$$\min |\Delta_V| \geq 3.04288, \quad \min |L^2 \Delta_V|_{\ell=2} \geq 109.54368,$$

and

$$\min (2L\alpha_4)_{\ell=2} \geq 0.119952.$$

D.3 Eight closure gates

The independent certificate returns `true` for all of the following tests:

- G1 positive f,h on all handover bands
- G2 monotone scalar patch coordinate ϕ_i
- G3 rank-three SVT Hessian-control map everywhere
- G4 monotone C-infinity-flat scalar partition
- G5 strict frozen pure-sector local stability margins
- G6 nonzero delta-A0 constraint pivot
- G7 nonzero H1/delta-A1 constraint determinant
- G8 nonzero common h1 pivot

and records the final status

PASS_CONSTRUCTIVE_LOCAL_SAME_ACTION_CLOSURE_CERTIFICATE

The reproducibility artifacts associated with this manuscript are

ssz_p5_same_action_closure_certificate_2026-09-14.json
 ssz_p5_same_action_closure_certificate_2026-09-14.py
 ssz_p5_same_action_closure_bands_2026-09-14.csv
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Internal SSZ/JIF research artifacts used for provenance and regression: `SSZ_CANONICAL_FORMULAS_2026.md`; *SSZ Variational Bridge Paper* (Wrede, Casu, Bingsi AI, 2026), `SSZ_Variational_Bridge_Paper_Wrede_Casu_Bi`; `SSZ_Middle_Bridge_Dynamics_Gap_Audit_2026-08-29.md`; and the 14 September 2026 P5 geometry/same-action closure certificates listed in Appendix D.

Appendix E. Extended source list

Primary project sources

1. `SSZ_P5_Horndeski_SVT_CLOSURE_COMPLETE_2026-09-14.md` — consolidated manuscript used as the main text basis for this PDF.
2. `ssz_p5_reconstructed_profile_2026-09-14.csv` — reconstructed frozen P5 background profile $((u, x, , f, h, Q, W))$.
3. `ssz_p5_geometry_keypoints_2026-09-14.csv` — key background points, including both light rings and handover locations.
4. `ssz_p5_light_ring_diagnostics_2026-09-14.csv` — ring positions and local potential diagnostics.
5. `ssz_p5_same_action_closure_certificate_2026-09-14.json` — certified closure gates, overlap bands, and pivot bounds.
6. `ssz_p5_same_action_closure_bands_2026-09-14.csv` — numerical summary of same-action overlap bands.

External physics references

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